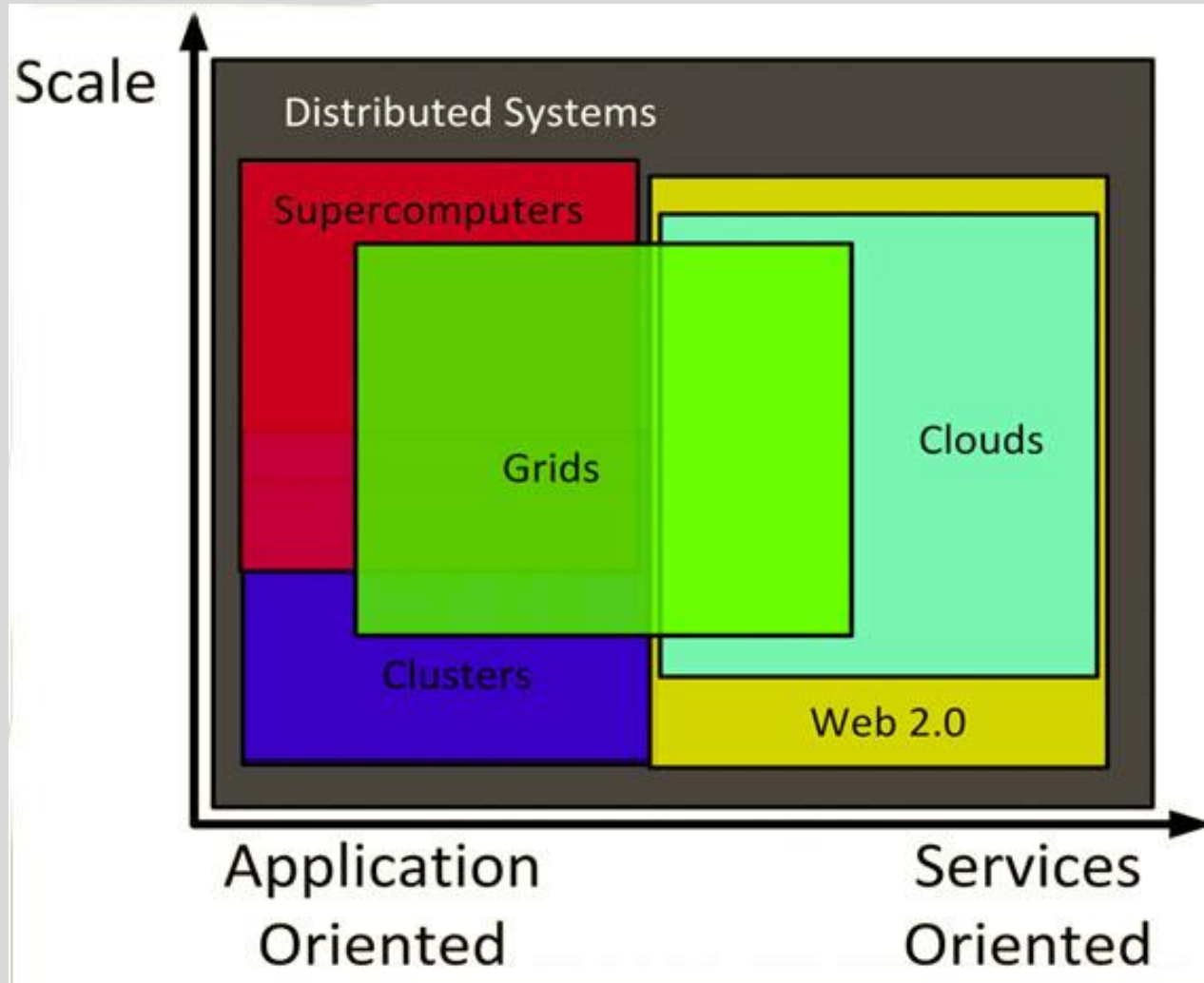


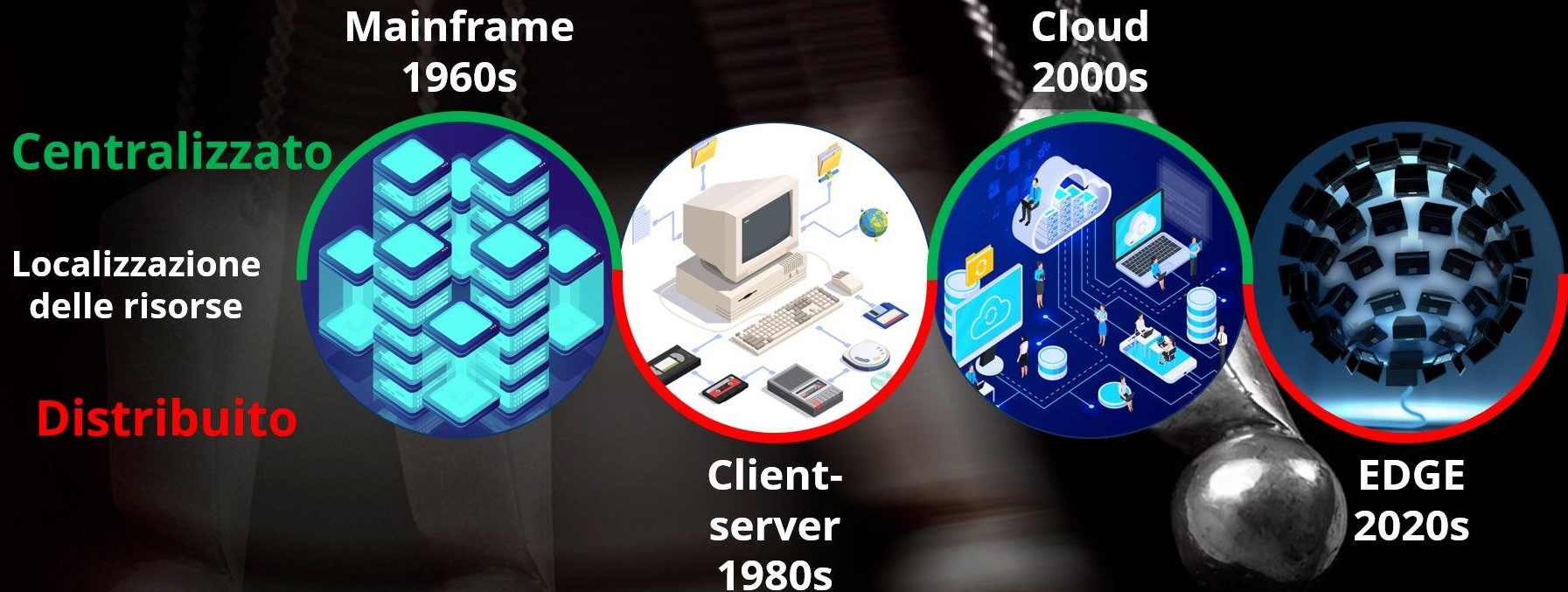
Amministrazione di Sistema

Cloud Computing

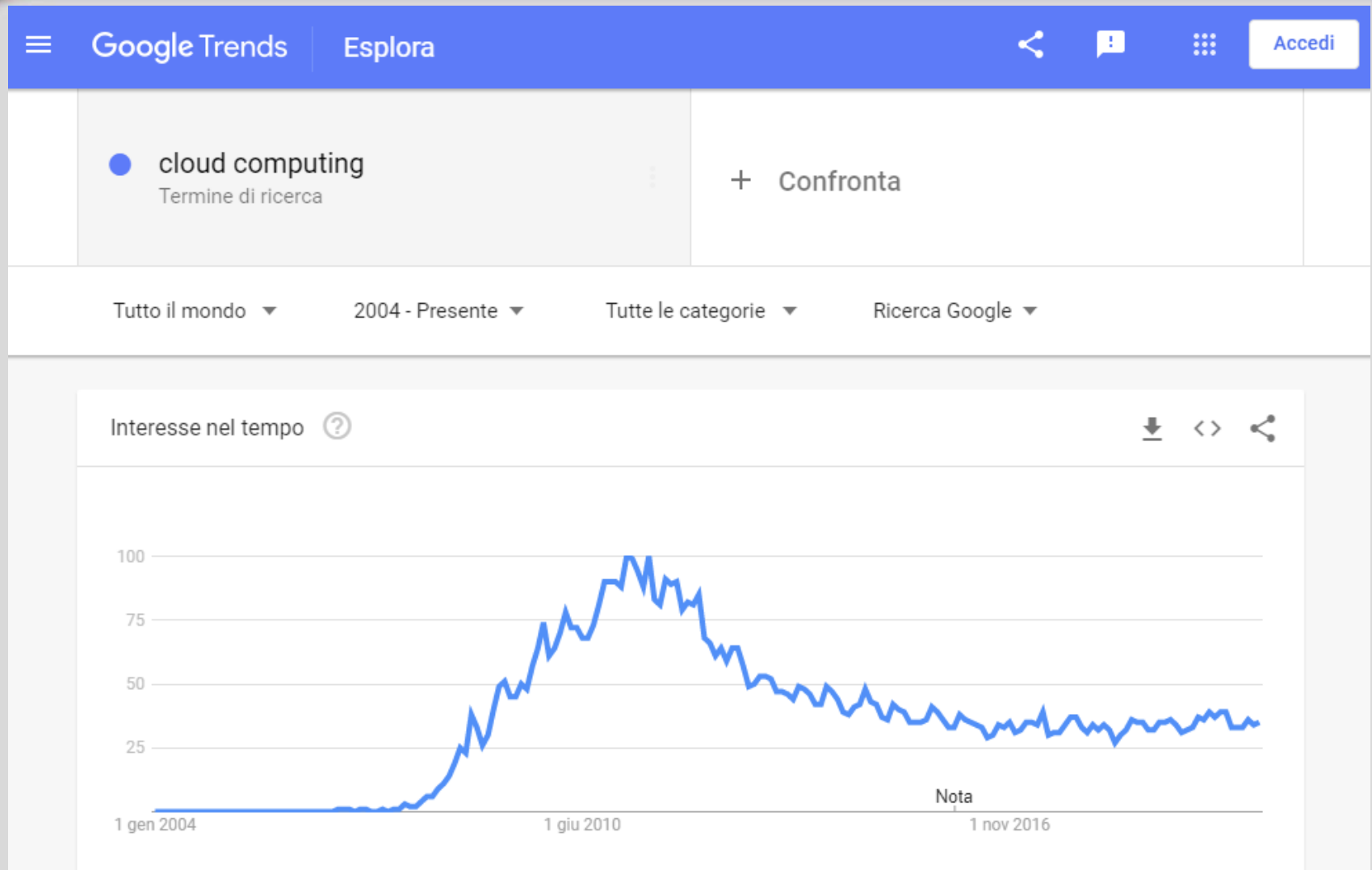
Classificazione



Il Pendolo dell'IT



Interesse verso il cloud



Cloud computing was popularized with [Amazon.com](https://www.amazon.com) releasing its [Elastic Compute Cloud](https://aws.amazon.com/ec2/) product in 2006

Cloud computing

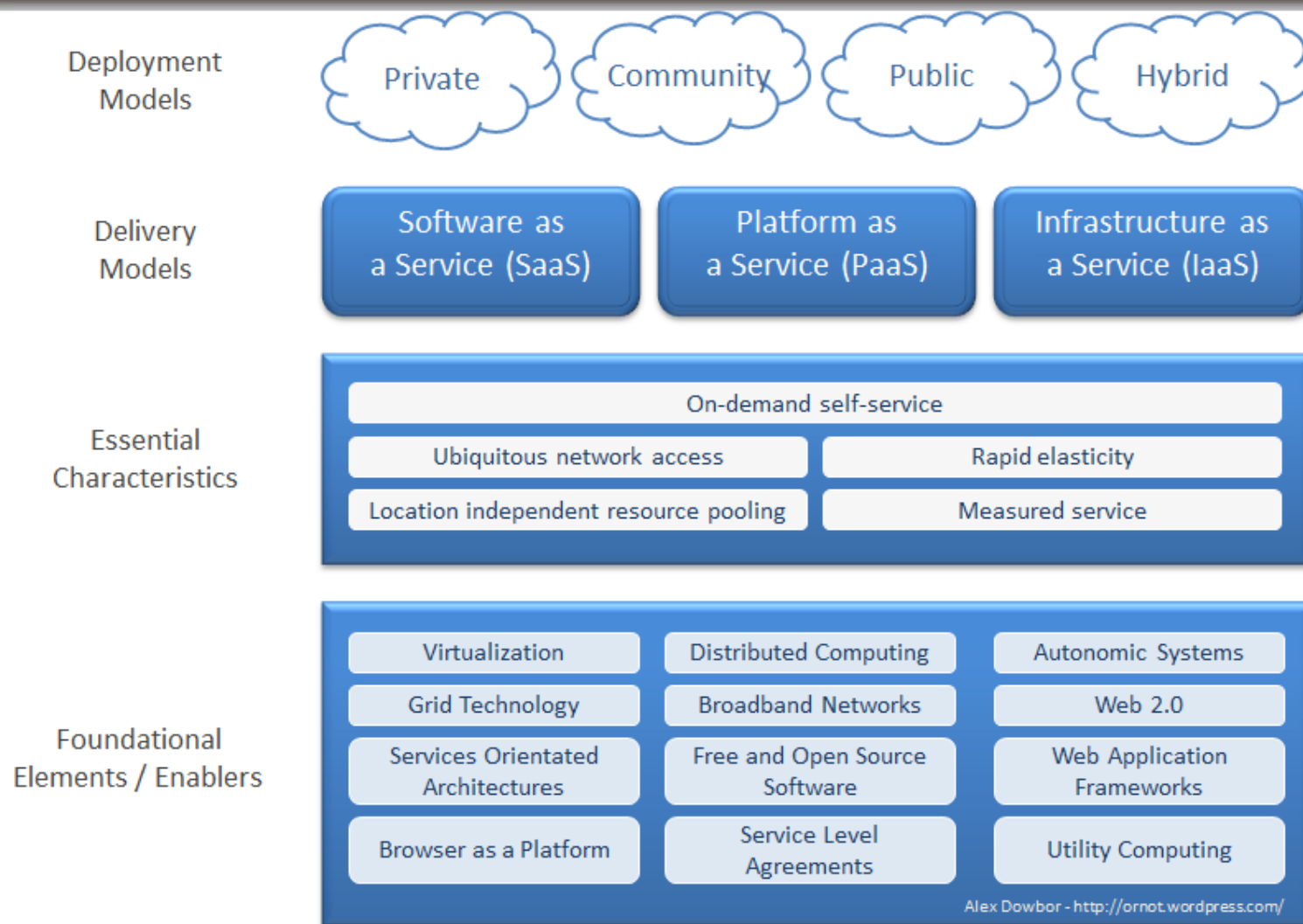
- Cloud computing
 - insieme di tecnologie informatiche che permettono l'utilizzo di risorse (storage, CPU) distribuite
 - sistemi distribuiti o (de-)centralizzati?
 - the provisioning of services in a timely (near on instant), on-demand manner, to allow the scaling up and down of resources
- Servizio
 - in modalità pay-per-use (ma non solo)
 - con self-provisioning (e anche API-driven)
 - con vasta scelta di servizi
- Cloud is not a destination (or place), it's an experience
- AAA (Astratto, Agile, Automatico)

Definizione



- NIST (National Institute of Standards and Technology)
- “Five essential characteristics”:
 - On-demand self-service
 - A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
 - Broad network access
 - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
 - Resource pooling
 - The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
 - Rapid elasticity
 - Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear unlimited.
 - Measured service
 - Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts).

NIST Cloud Computing Model



Based on the NIST Working Definition of Cloud Computing v14 and
<http://www.csrc.nist.gov/groups/SNS/cloud-computing/index.html>

Creative Commons Attribution-Share Alike 3.0
Alexander Dowbor - <http://ornot.wordpress.com>

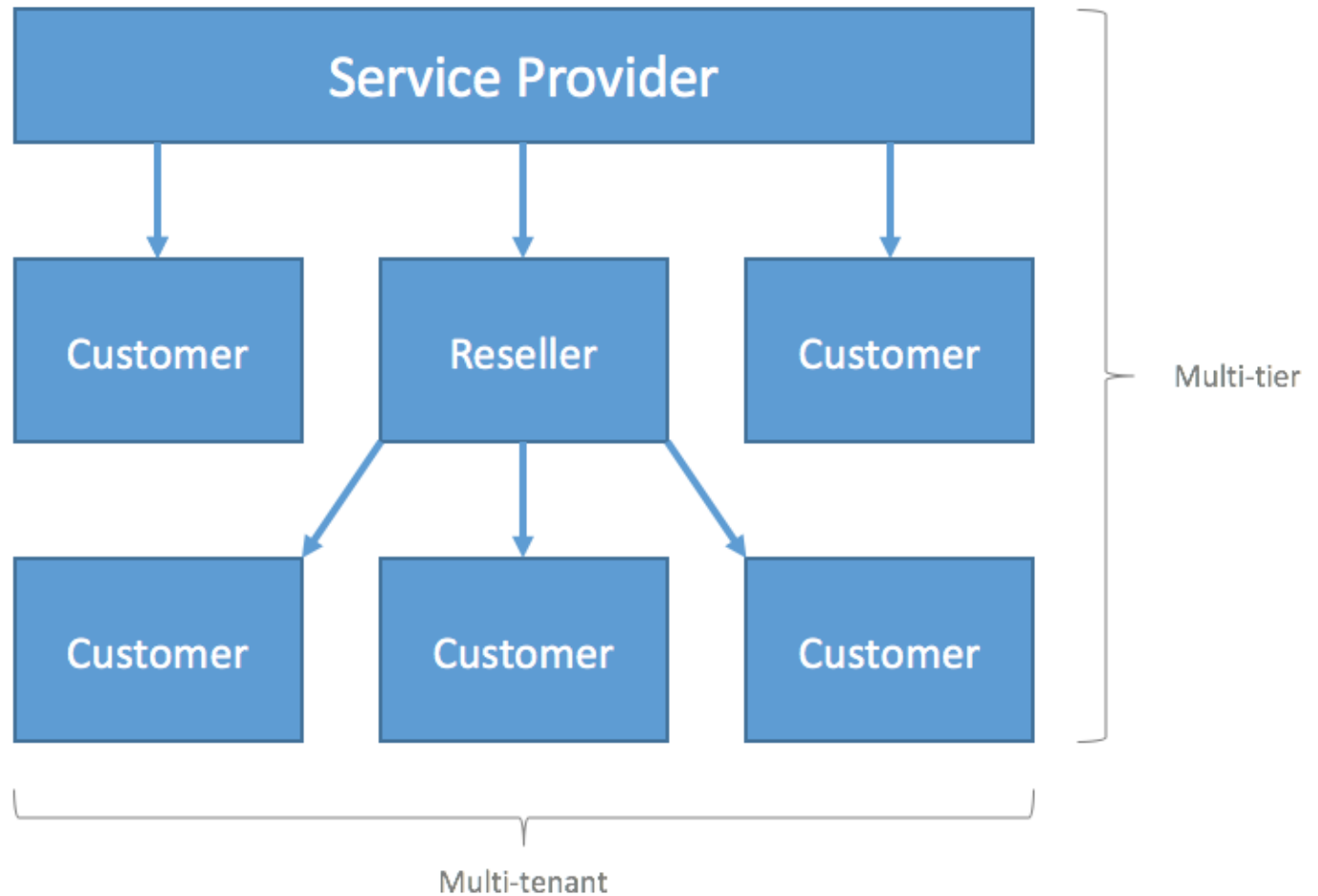
To Qualify as a Cloud

- **Common, Location-independent, Online Utility on Demand***
 - **Common** implies multi-tenancy, not single or isolated tenancy
 - **Utility** implies pay-for-use pricing
 - **on Demand** implies ~infinite, ~immediate, ~invisible scalability
- Alternatively, a “Zero-One-Infinity” definition:**
 - 0** On-premise infrastructure
 - Acquisition cost
 - Adoption cost
 - Support cost
 - 1** Coherent and resilient environment – not a brittle “software stack”
 - ∞** Scalability in response to changing need
 - Integratability/Interoperability with legacy assets and other services
 - Customizability/Programmability from data, through logic, up into the user interface without compromising robust multi-tenancy

* Joe Weinman, Vice President of Solutions Sales, AT&T, 3 Nov. 2008

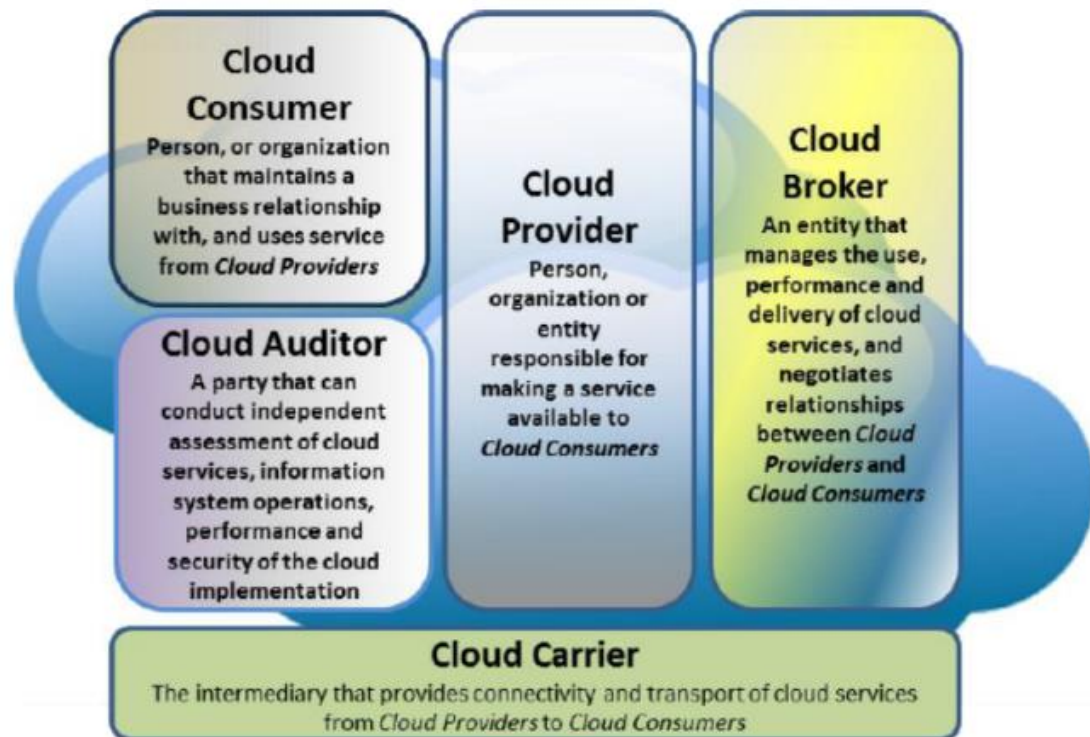
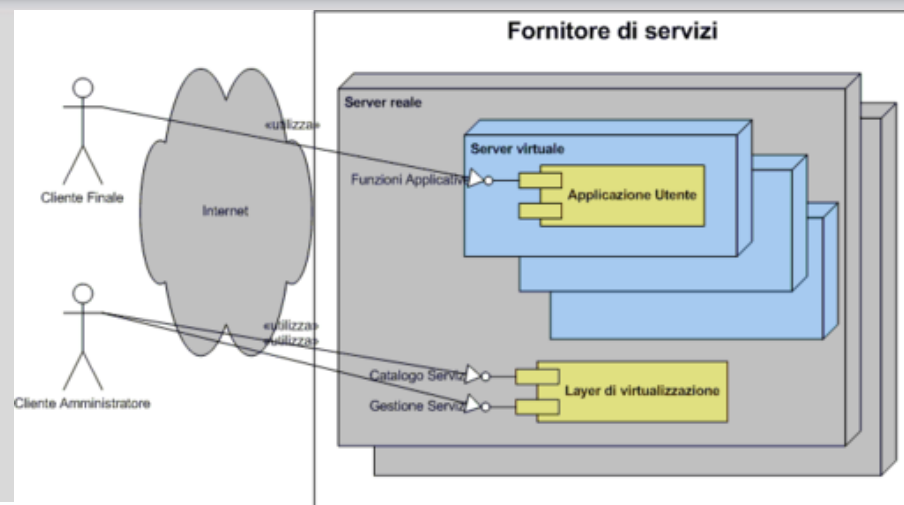
** From The Jargon File: “Allow none of foo, one of foo, or any number of foo”

Multi-tenant Multi-tier



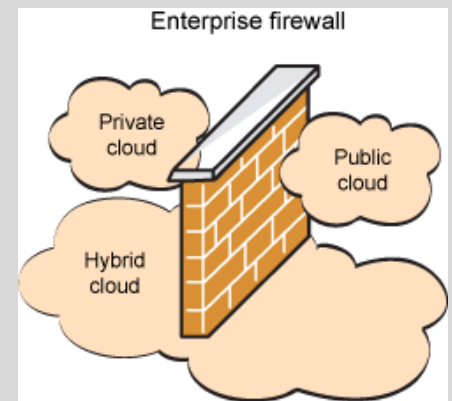
Ruoli nel cloud

- Definizione di due ruoli
 - Provider
 - Consumer (user)
 - Tenant admin
 - ...



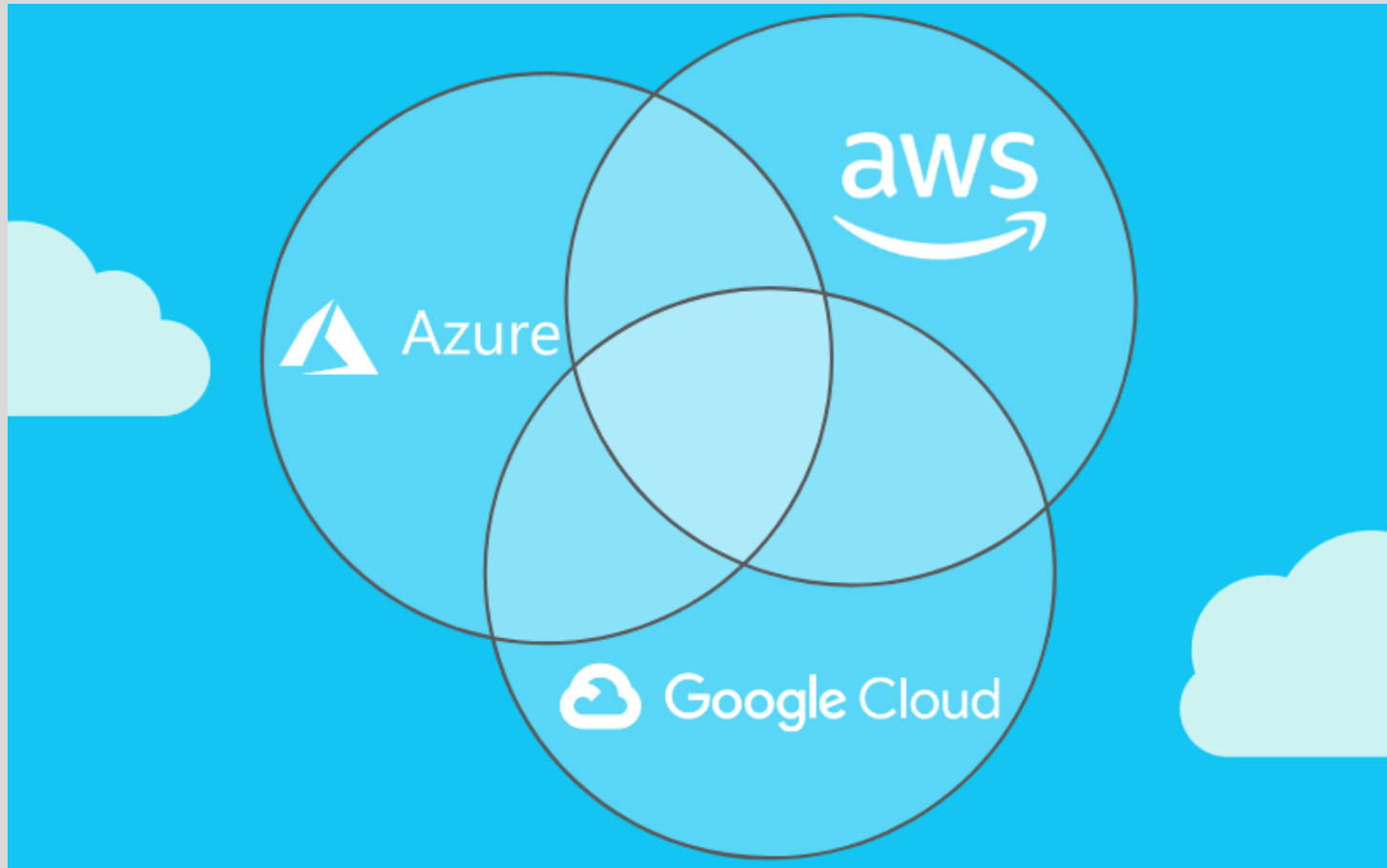
Tipi di cloud

- Il cloud non è un posto ma un modello
- Ma può essere classificato in base alla modalità di accesso ai servizi
- Deployment models:
 - Public (External) Cloud
 - Private (On-Premises) Cloud
 - Hybrid Cloud
 - Community Cloud



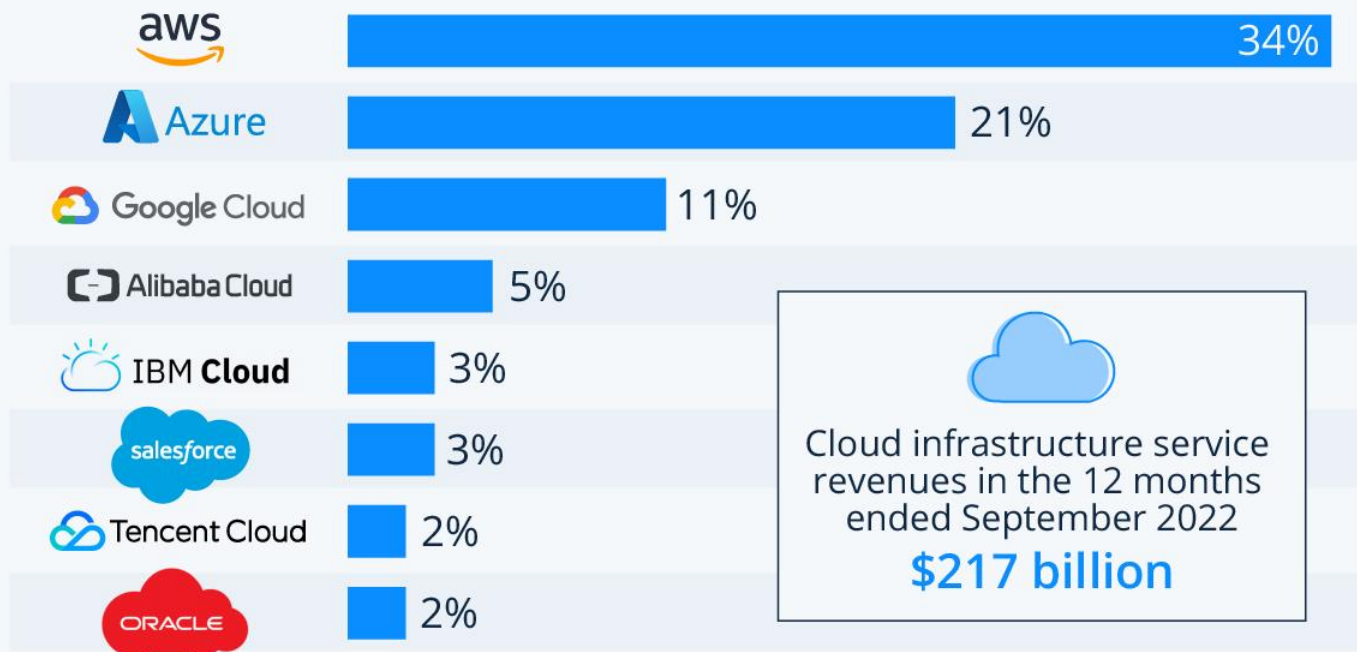
Public Cloud

- Provider mondiali più importanti



Amazon, Microsoft & Google Dominate Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q3 2022*



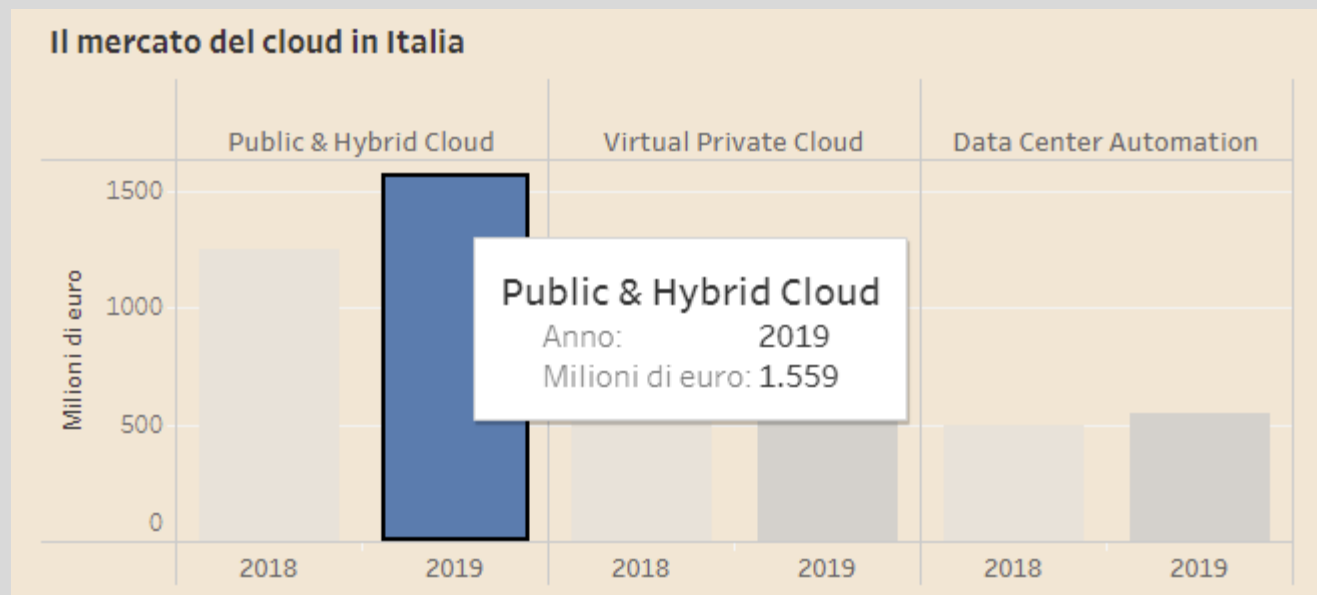
* includes platform as a service (PaaS) and infrastructure as a service (IaaS) as well as hosted private cloud services

Source: Synergy Research Group



Public Cloud in Italia

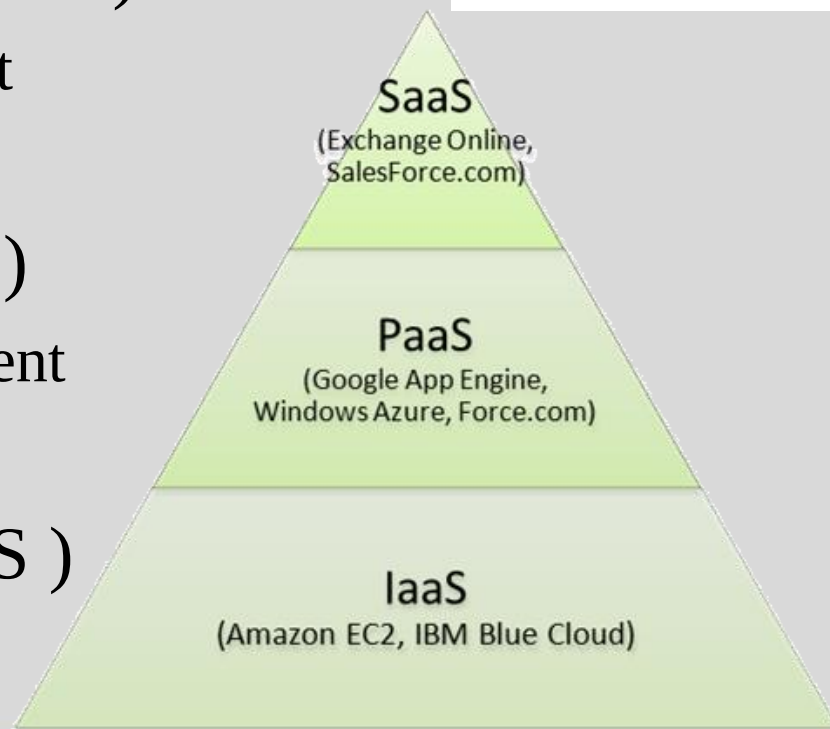
- AWS Region
- Azure Region
- Cloud.it
- Noovle + Google



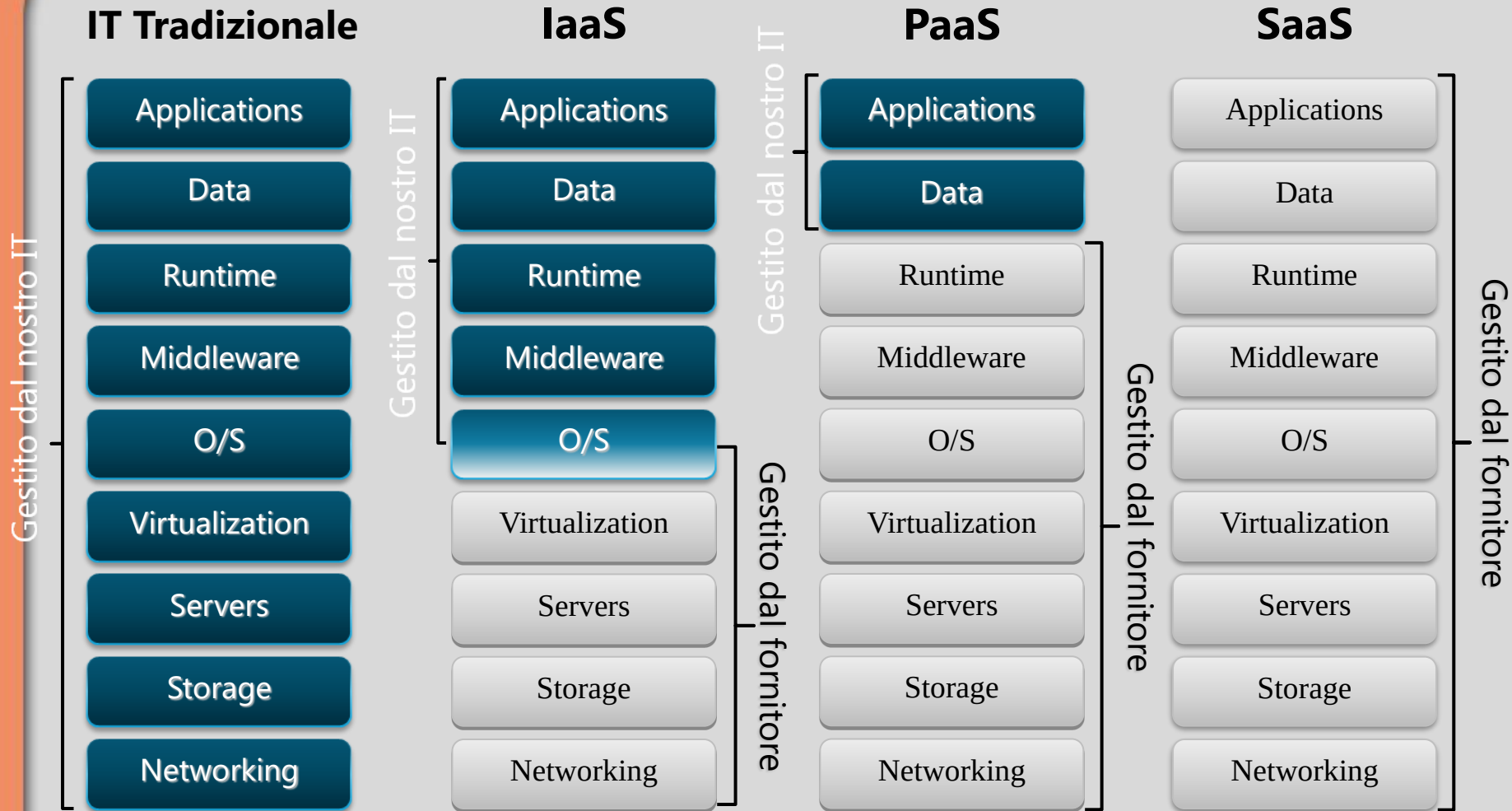
Tipi di servizi cloud

Client
Application
Platform
Infrastructure
Server

- Software as a service (SaaS)
 - Best practices under development
 - Es: Microsoft 365, Google Workspace
- Infrastructure as a service (IaaS)
 - Standards under development
 - Es: VM
- Platform as a service (PaaS)
 - Technology under development
 - Es: WebApps, DBaaS
- Hardware as a service (HaaS)
- All as a Service (XaaS)



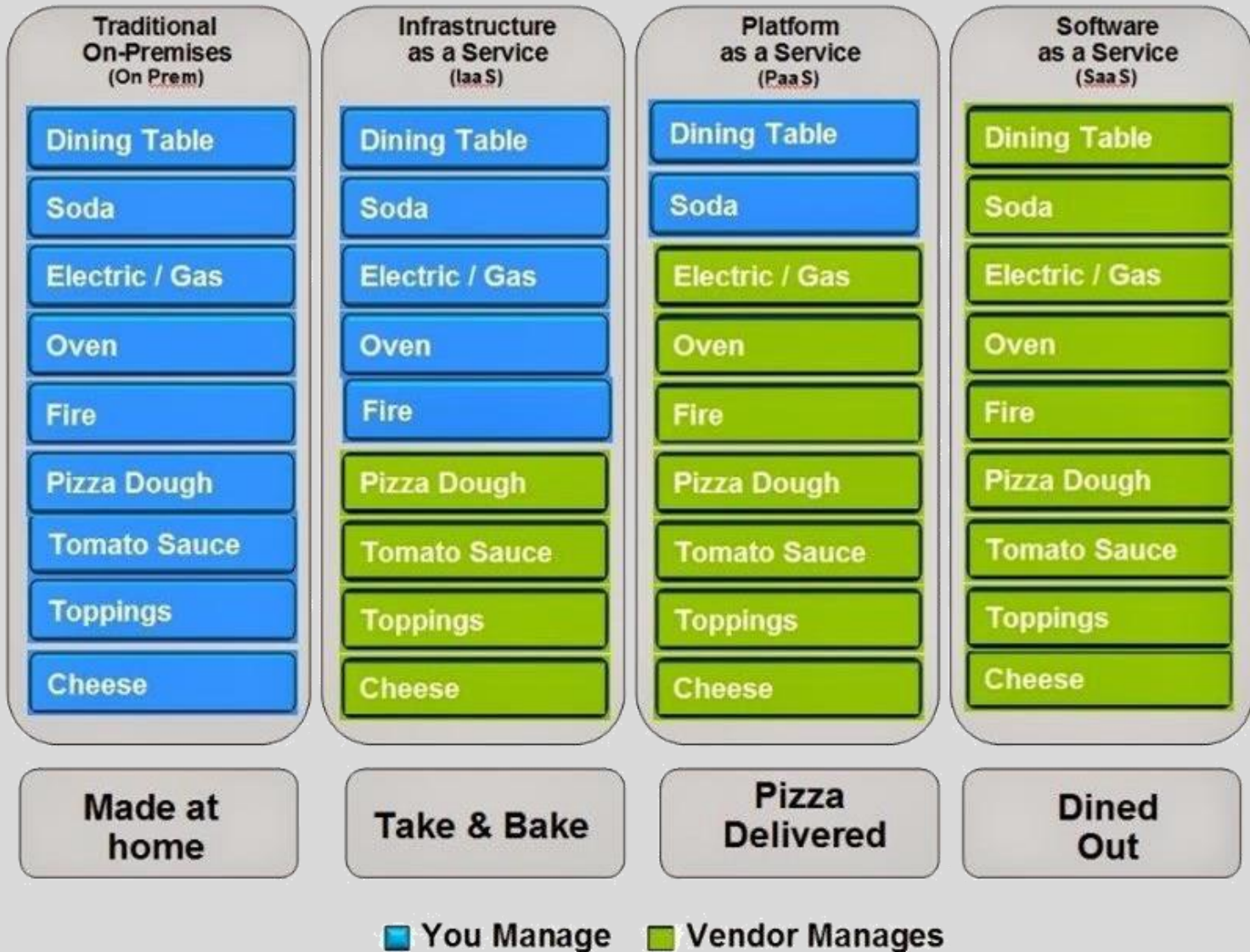
Tassonomia del cloud



Evoluzione dei servizi

- CaaS – Container as a Service
- KaaS – Kubernetes as a Service
- DBaaS – Database as a Service
- BaaS – Backup as a Service
- DRaaS – Disaster Recovery as a Service
- ...

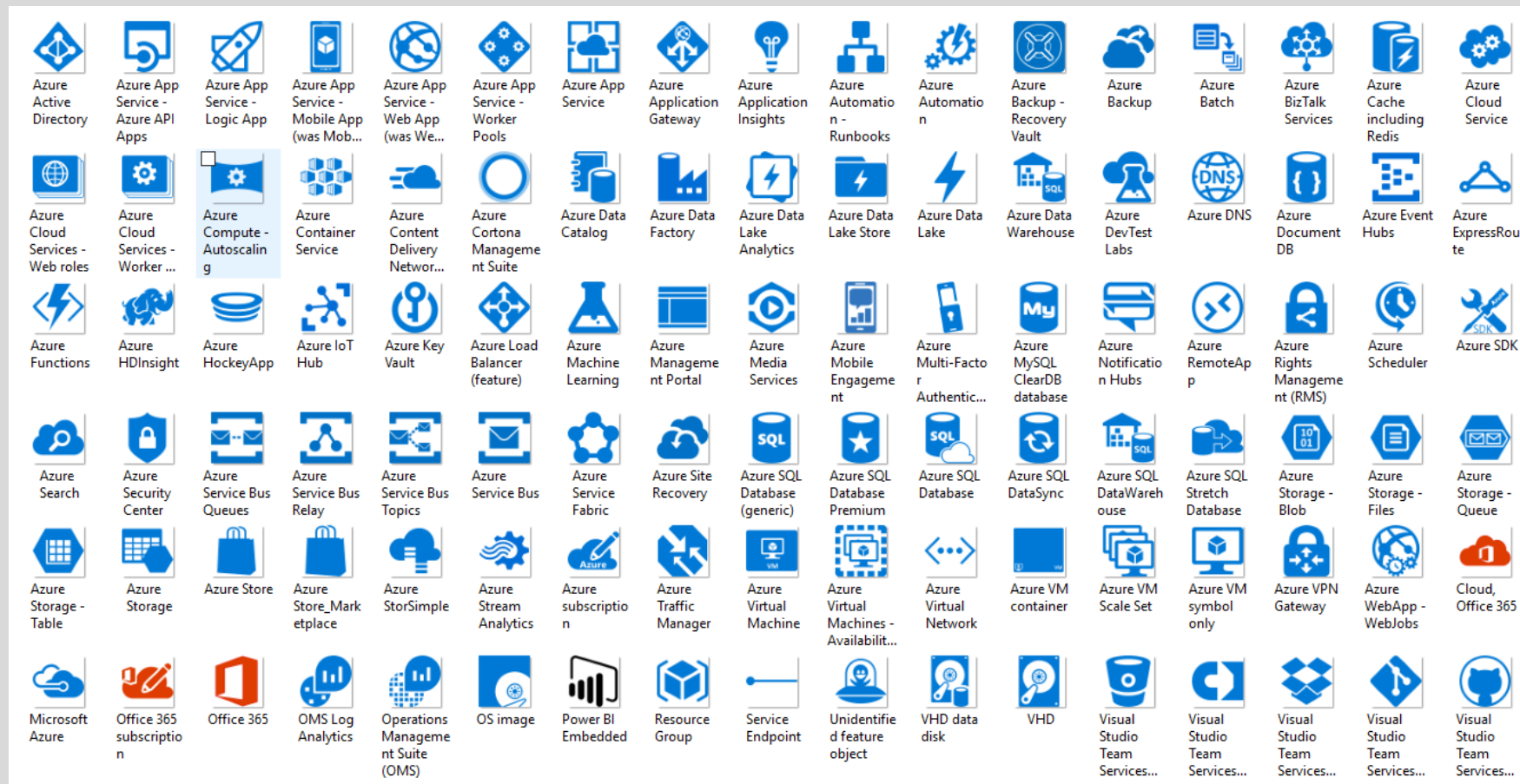
Pizza as a Service



Tipi di servizi

Services	GCP	Azure	AWS
IaaS	Google Compute Engine	Azure VM	Amazon Elastic Compute Cloud (EC2)
PaaS	Google App Engine	App Service and Cloud Services	AWS Elastic Beanstalk
Containers	Google Kubernetes Engine	Azure Kubernetes Service (AKS)	Amazon Elastic Container Service
Serverless (FaaS)	Google Cloud Functions	Azure Functions	AWS Lambda
DBaaS	Cloud SQL	Azure DB	AWS Relational Databases (RDS)
Object Storage	Blob Storage	Blob Storage	Simple Cloud Storage (S3)

Rich set of services



Servizi e Provider

Service category ▼	Service type	Google Cloud product	Google Cloud product description	AWS offering	Azure offering
App Modernization	CI/CD	Cloud Build	Build, test, and deploy on Google Cloud serverless CI/CD platform	AWS CodeBuild, AWS CodeDeploy, AWS CodePipeline	Azure DevOps, Github Enterprise
App Modernization	Execution Control	Cloud Tasks	Control and observe asynchronous service requests between independent applications using this zonal, execution-control service.	Amazon Simple Queue Service (SQS), Amazon Simple Notification Service (SNS)	Azure Service Bus, Azure Storage Queues
App Modernization	Multi-cloud	Anthos	Anthos is a managed application platform that extends Google Cloud services and engineering practices to your environments so you can modernize applications faster and establish operational consistency across them.	Amazon EKS Anywhere, Amazon ECS Anywhere	Azure Arc
App modernization	Multi-cloud	Anthos Clusters	Extend GKE to work in multiple environments, including attached clusters, AWS, Azure, bare metal, and VMWare.	Amazon EKS Anywhere	
App modernization	Multi-cloud	Anthos Config Management	Automate policy and security at scale for your hybrid and multi-cloud Kubernetes deployments.	Chef Automate AWS OpsWorks	Azure App Configuration
App modernization	Multi-cloud	Config Connector	Manage Google Cloud resources through Kubernetes.	AWS Controllers for Kubernetes	Azure Service Operator
App modernization	Multi-cloud	Container-Optimized OS	Efficiently and securely run Docker containers on Compute Engine VMs.	AWS Bottlerocket	Azure Container Instances
App modernization	Multi-cloud	Hybrid Connectivity	Connect your infrastructure to Google Cloud on your terms, from anywhere.	AWS Direct Connect	Hybrid Connections

Dark side of the cloud

Ransomware-as-a-Service is now an Industry with Large Payouts

Security Landscape



Attackers partner to cover different aspects of the attack and get % of payout (business model)

Initial access and credential compromise sellers

Experienced Pen testers provide the exploit

Ransomware Dev provide the payload

Operating as pipeline

Double and triple extortion since mid-2020

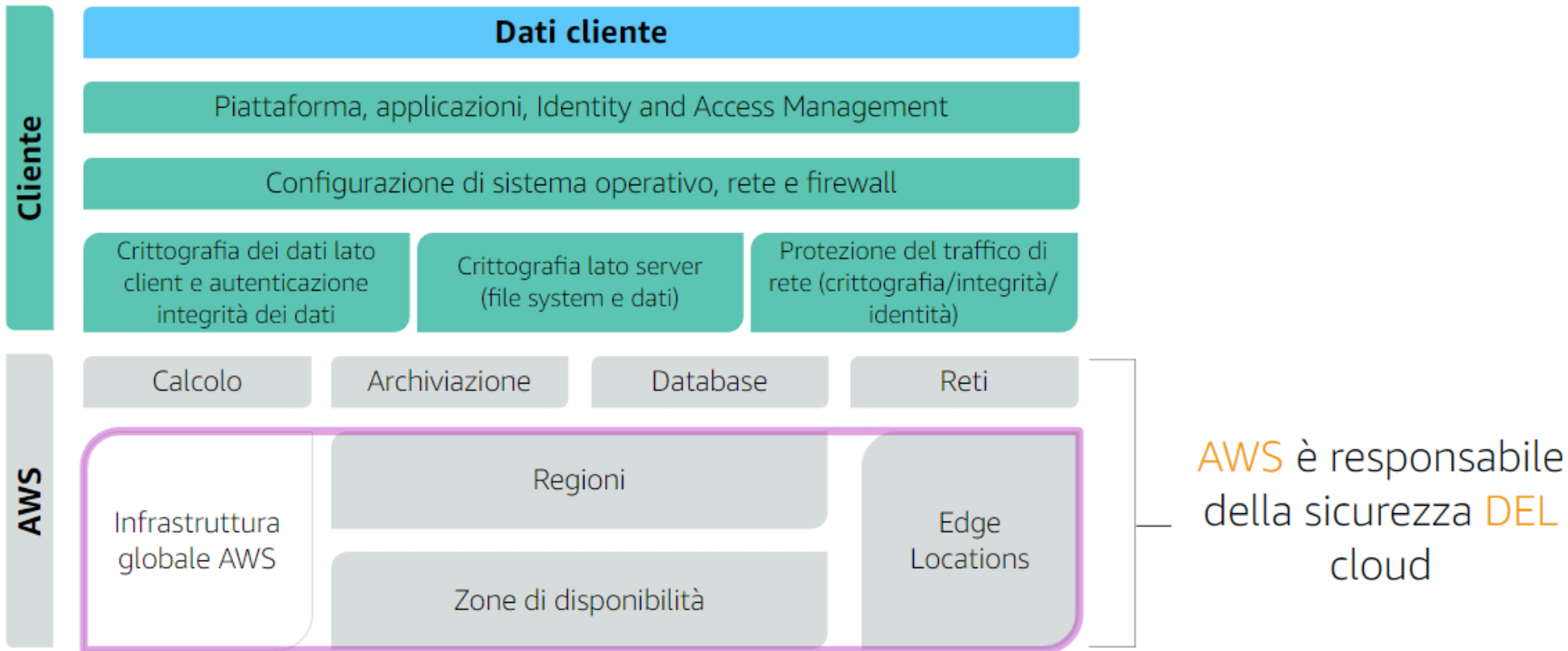
Shared responsibility

Shared responsibility model

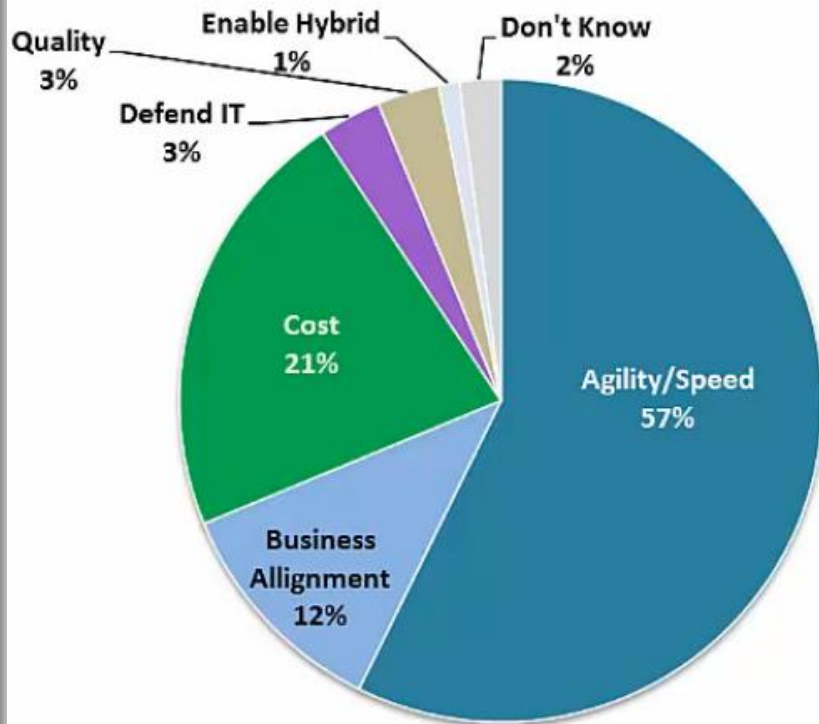


<https://docs.microsoft.com/it-it/azure/security/fundamentals/shared-responsibility>

Shared responsibility



Perché scegliere il cloud?



Source: Gartner -The Drivers and Challenges of Private Cloud Computing, Mar 2012

INCREASE BUSINESS AGILITY

- Enable the business to quickly react to market demands
- Capture market share

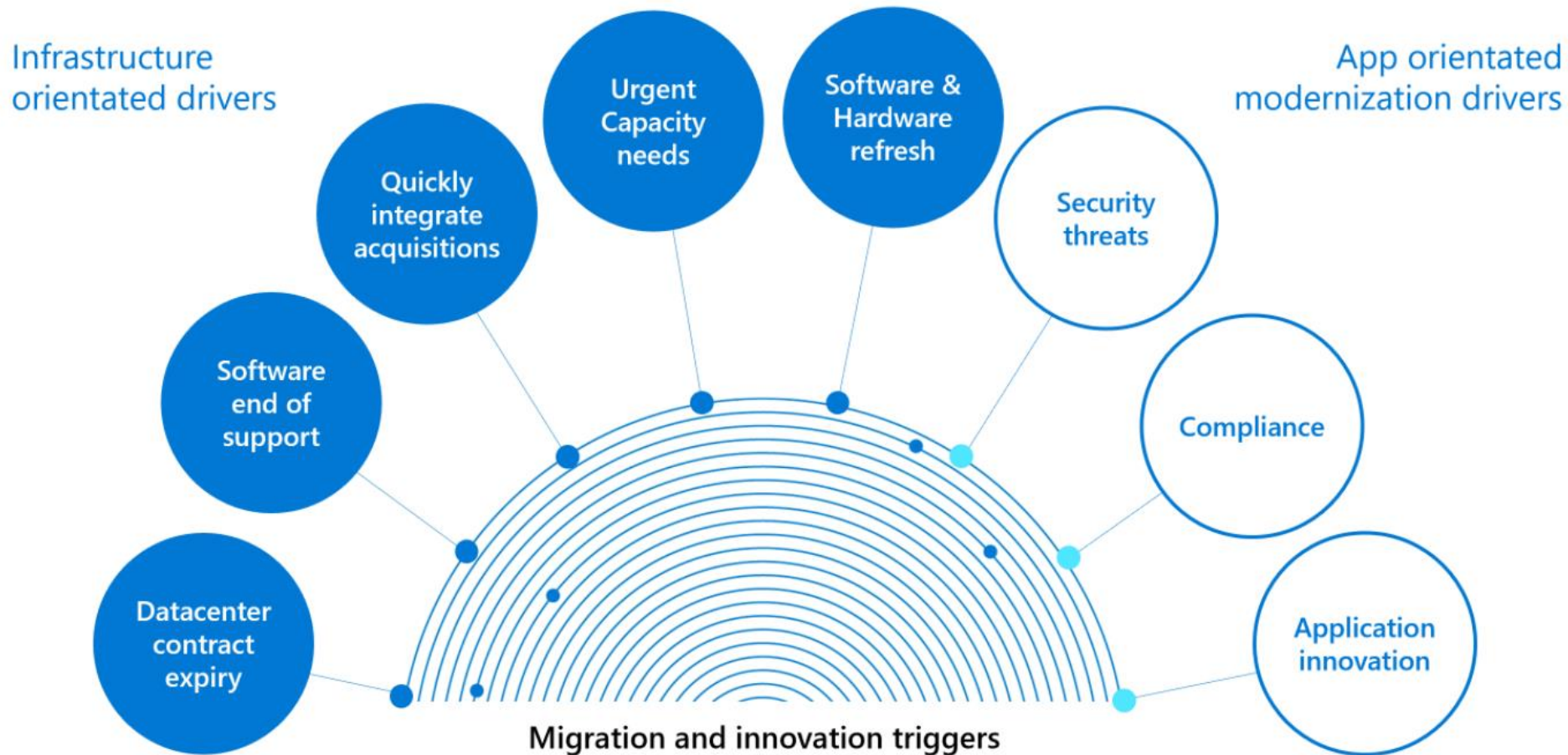
IMPROVE IT EFFICIENCY

- Improve productivity through self-service & automation
- Optimize resource utilization

CONSUMERIZATION OF IT

- Improve customer satisfaction
- Secure IT relevancy to business

Perché scegliere il cloud?

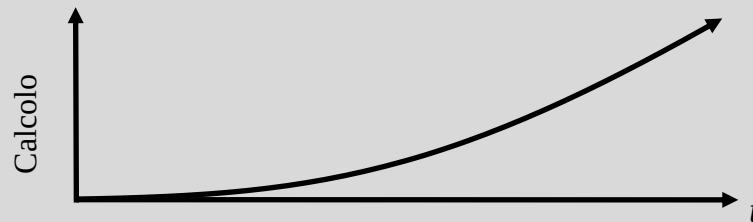


Quando conviene il public cloud?



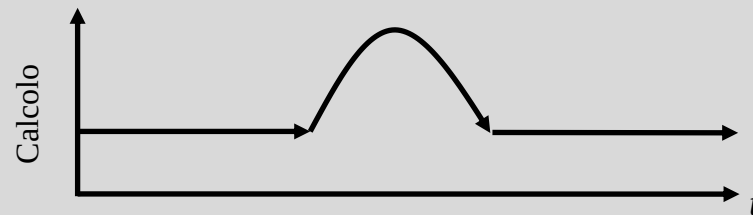
On & Off

Attività di tipo batch
In caso di eccesso di capacità inutilizzata
In caso di difficoltà a pianificare l'impiego di risorse



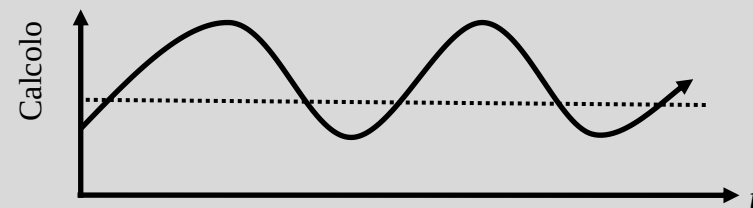
Crescita rapida

In caso di servizi che crescono rapidamente
In caso di nuove esigenze di business
Quando l'hardware non arriva in tempo utile



Carichi improvvisi

Quando si verificano condizioni imprevedibili
In caso di situazioni con picchi improvvisi
Copertura dei carichi di lavoro estremi

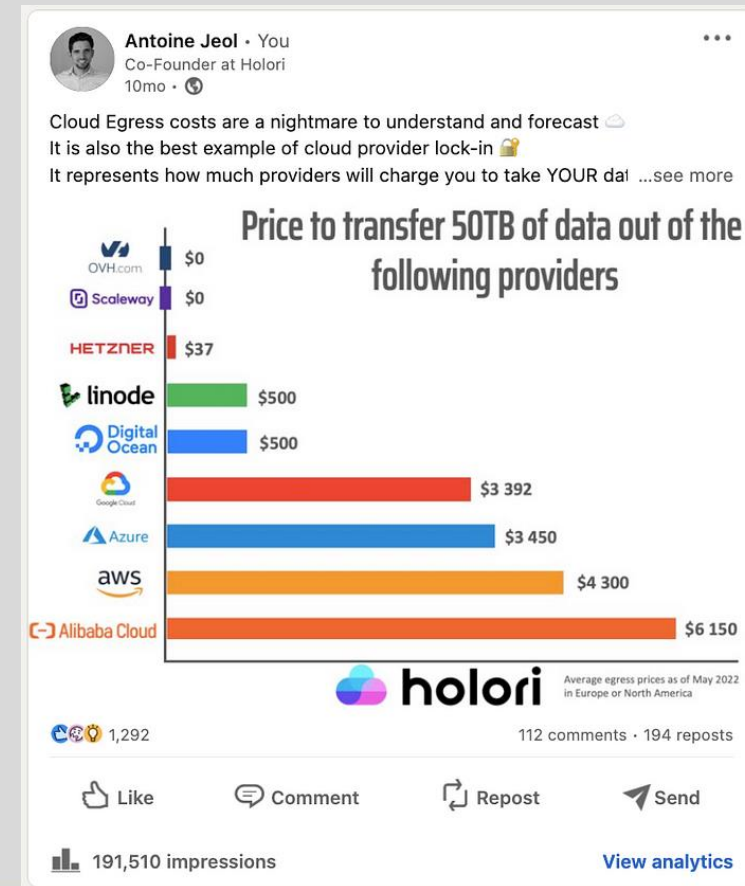


Carichi molto differenziati

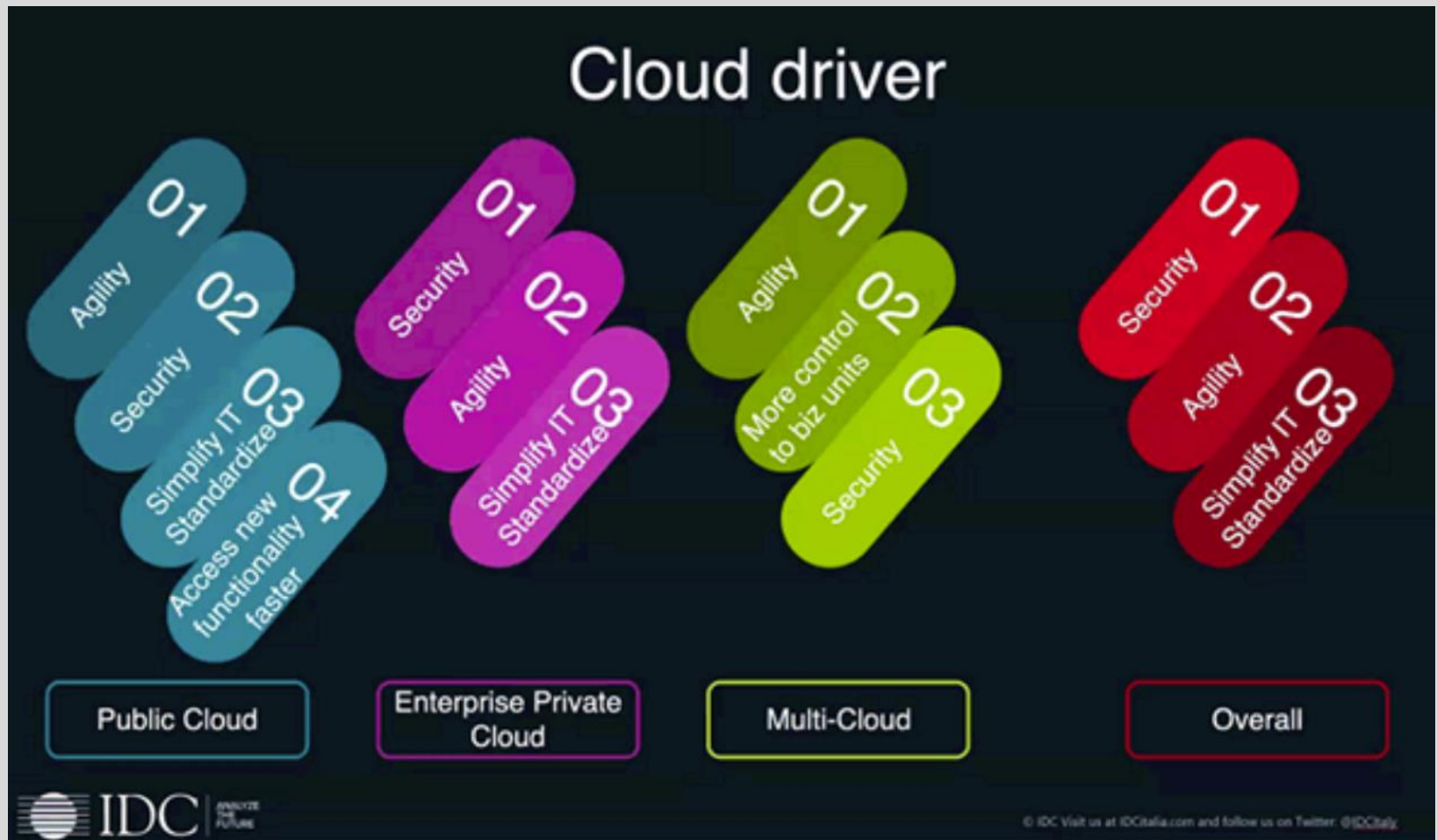
In caso di carichi di lavoro ad andamento ciclico
In caso di picchi di carico periodici
Quando non si riescono a bilanciare le risorse

Costi del public cloud

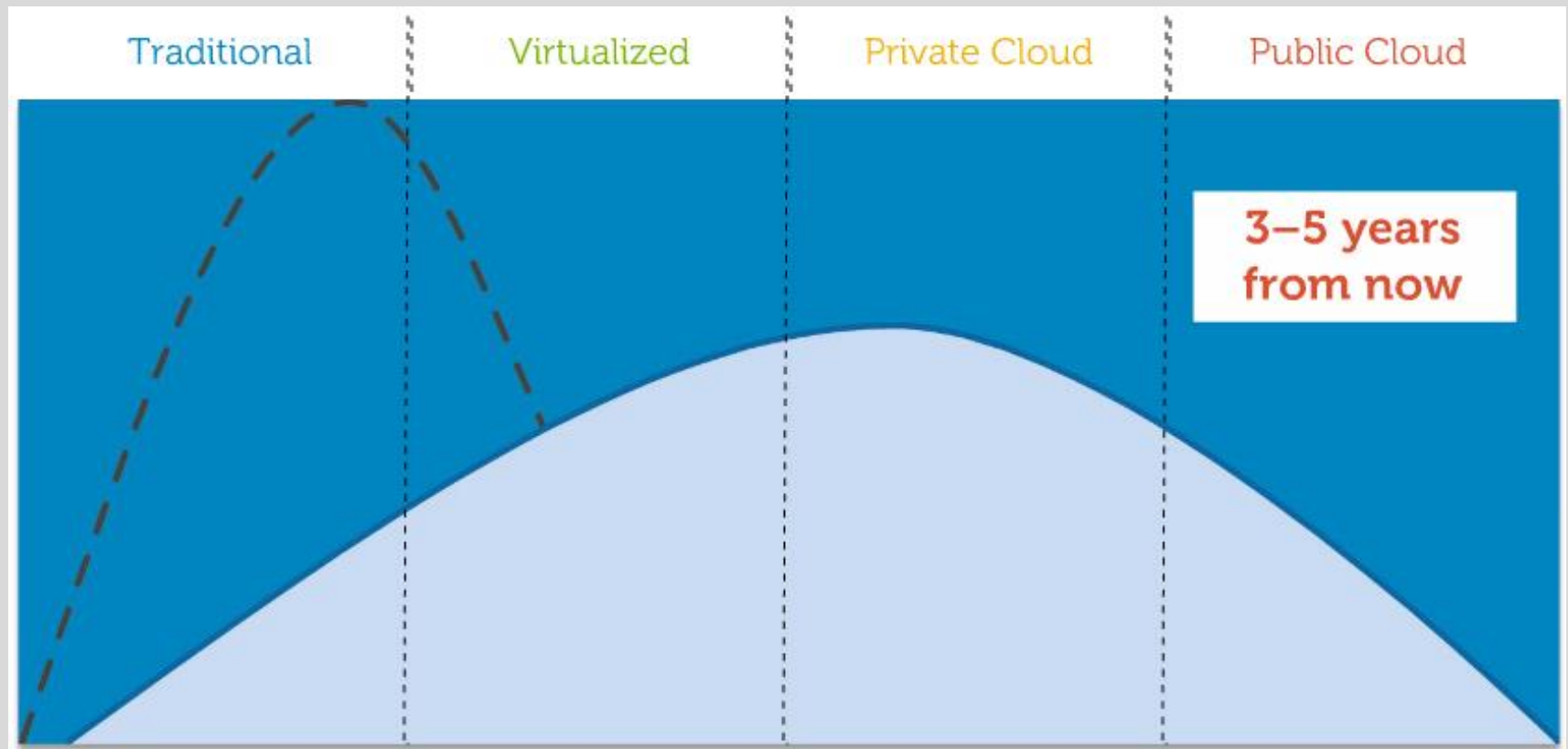
- È sempre conveniente?
- Costi rapportati al tipo di servizio
- Vari tipi di sconti
 - Risorse riservate
 - Risorse spot
 - ...
- Costi «nascosti»
 - IP pubblici
 - Storage
 - Banda (di download)
 - Auto-Scaling



Cloud driver



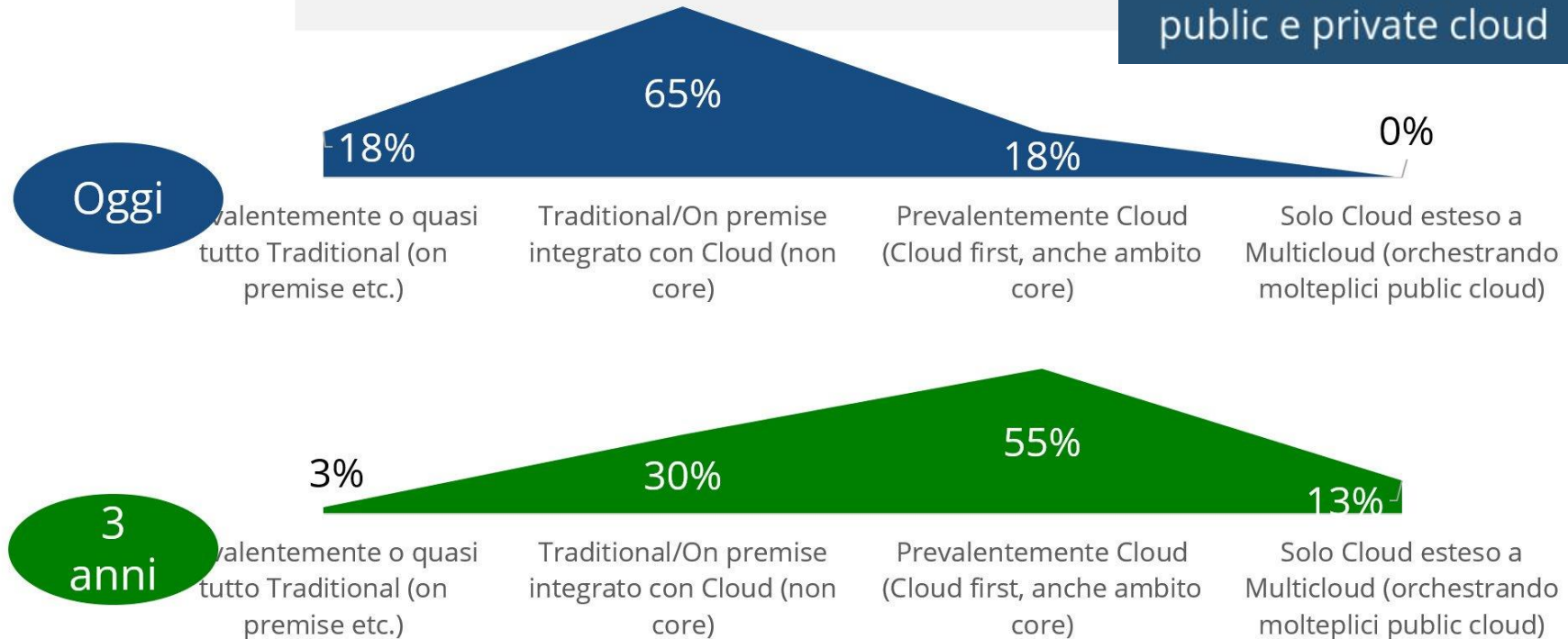
Come cambiano gli scenari



Verso il cloud first

Nel 2020 saranno spesi in Europa 64 miliardi di dollari in servizi di public e private cloud

Cloud: transizione inevitabile



Crescita del cloud

Worldwide Public Cloud Services Revenue and Year-Over-Year Growth 2019
(revenues in US\$ millions)

Segment	2018 Revenue	2017 Revenue	Year-over-Year Growth	Share of Public Cloud Market
IaaS	\$35,975	\$24,810	45.0%	22.1%
PaaS	\$25,257	\$18,050	39.9%	15.5%
SaaS (including SIS)	\$121,306	\$100,383	20.8%	62.4%
Total	\$182,538	\$143,242	27.4%	100.0%

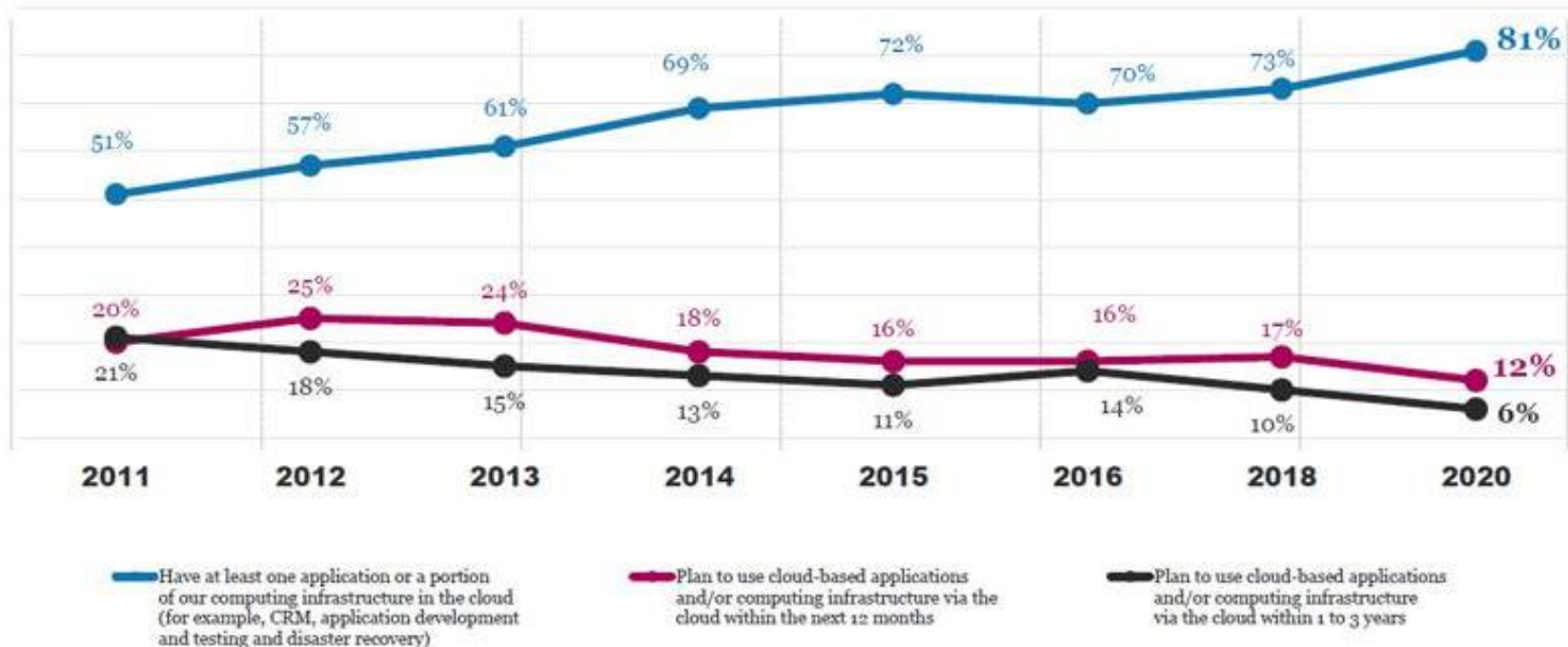
Source: IDC Worldwide Semiannual Public Cloud Services Tracker, 2H18



Crescita del cloud

- 32% Of IT Budgets Will Be Dedicated To The Cloud By 2021
- 81% of enterprises have at least one application or a segment of their computing infrastructure in the cloud today

Growth in Cloud Adoption



2020 Cloud Computing Study

- <https://www.idg.com/tools-for-marketers/2020-cloud-computing-study>
- 92% of organization's IT environment is at least somewhat in the cloud today, as only 8% say their total IT environment is all on-premises.
- There are dollars to be spent on cloud tools as close to one-third (32%) of total IT budget will be allocated to cloud computing within the next year.
- Cloud-first applications may become more prevalent over time – currently 46% of organization's cloud-based applications were purpose built for the cloud as 54% moved from an on-premises environment.
- More than half (55%) of organizations currently use multiple public clouds, with 21% saying they use three or more.
- The biggest challenges that ITDMs face when it comes to the ability to take full advantage of their public cloud resources are controlling cloud costs and data privacy/security challenges.

Cloud pubblico vs. on-prem



Michael Dell • Già segui

Founder, Chairman and CEO @Dell Technologies | Hardware, Softwar...
1s •

Not surprising:

83% of enterprises plan to move workloads back to private cloud ... altro

Mostra traduzione



The Future of the Enterprise Is Private

Private
Cloud



Private
AI

Private



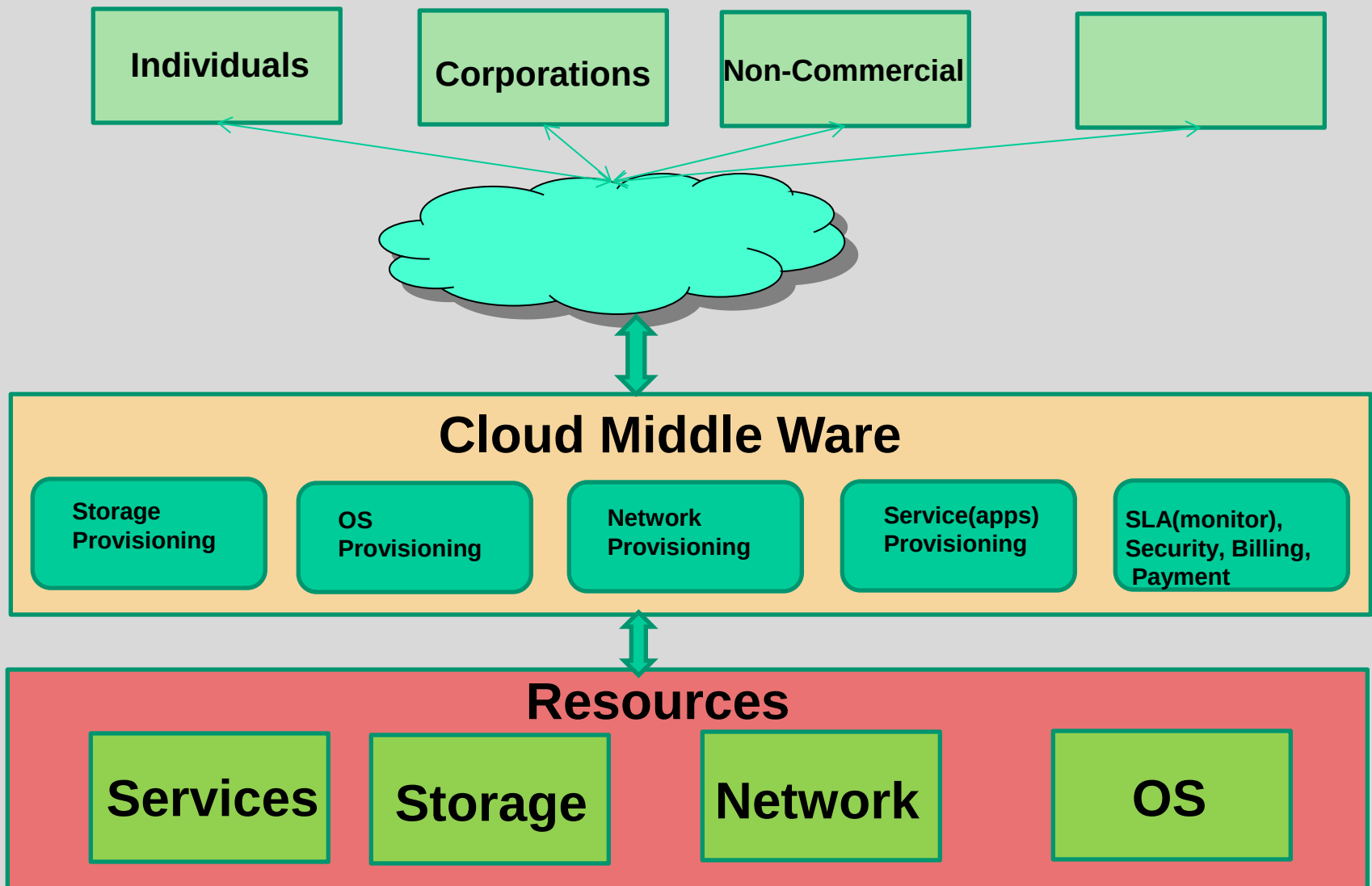
Data

Broadcom's Strategic Shift: VMware's Return to Private Cloud Emphasized at VMware Explore 2024 Las Vegas

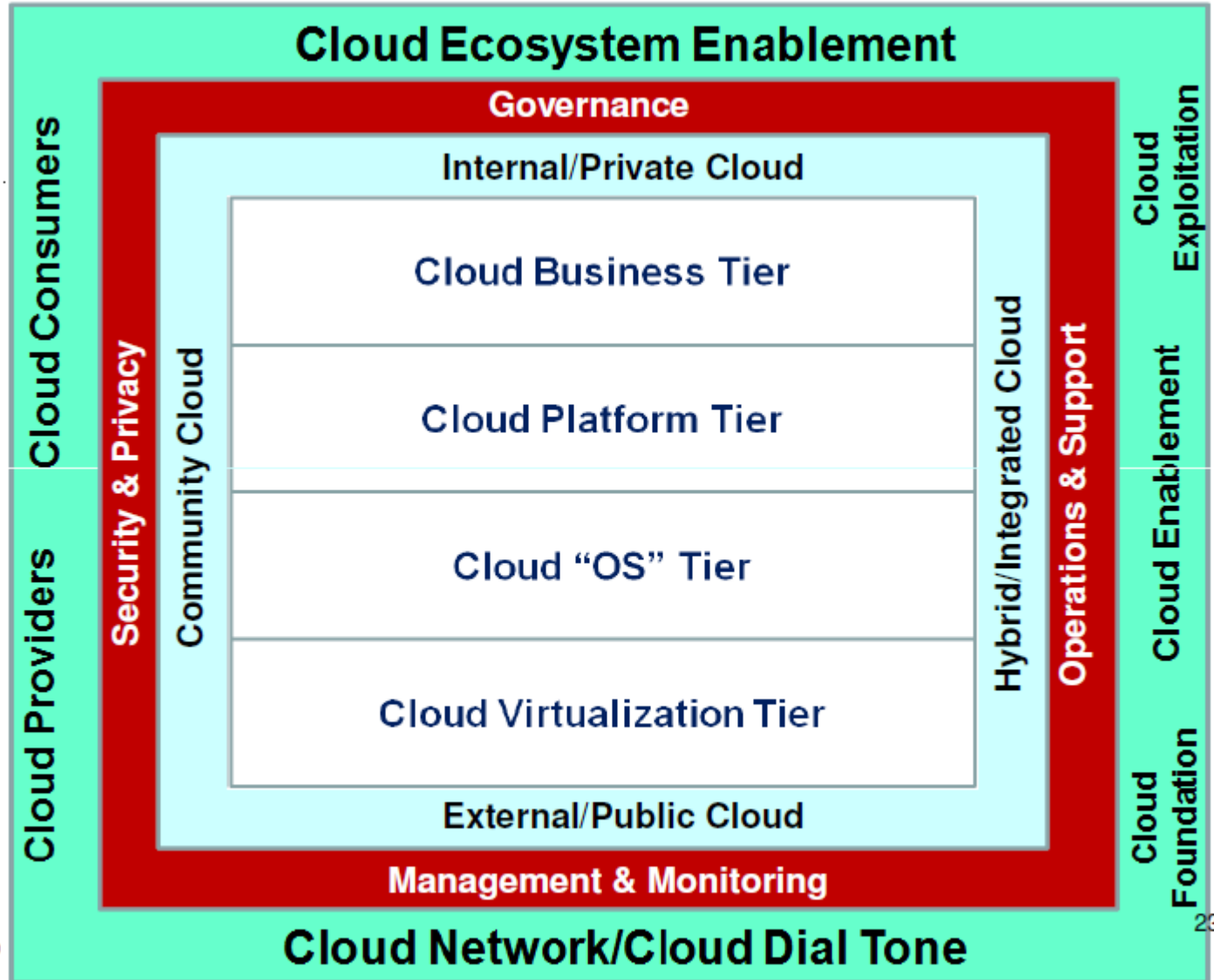
AAA nel cloud

- Abstraction
 - Automation
 - Agility
-
- Virtualizzazione vs. Cloud

What is a Cloud?

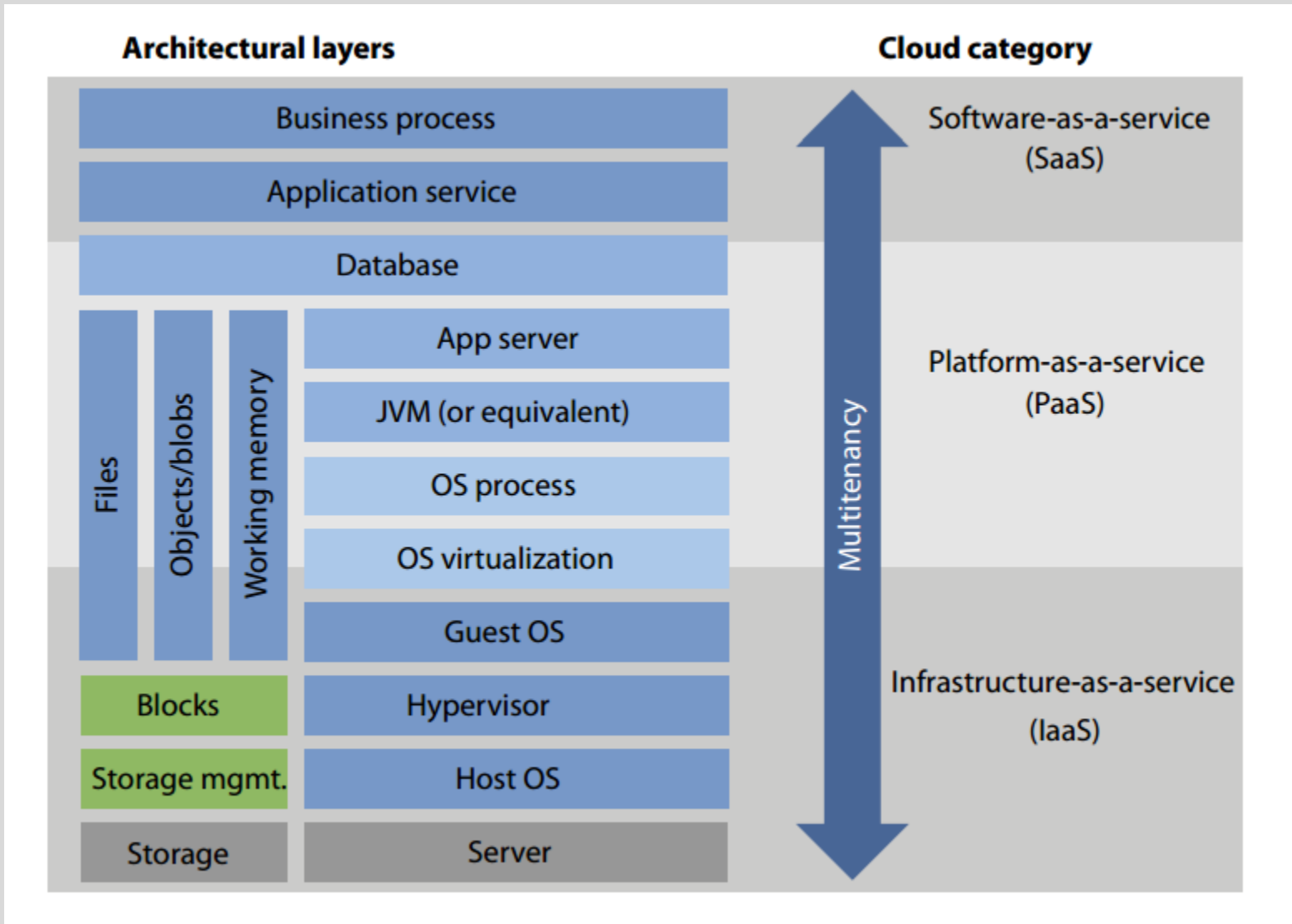


Più o meno complicato?



23

Più o meno complicato?



Regions & Availability zones

Web
App
DB

Web
App
DB

Web
App
DB

Region

Region

Availability Zone

Availability Zone

Availability Zone

Availability Zone

Data Center

Data Center

Data Center

Data Center

Data Center

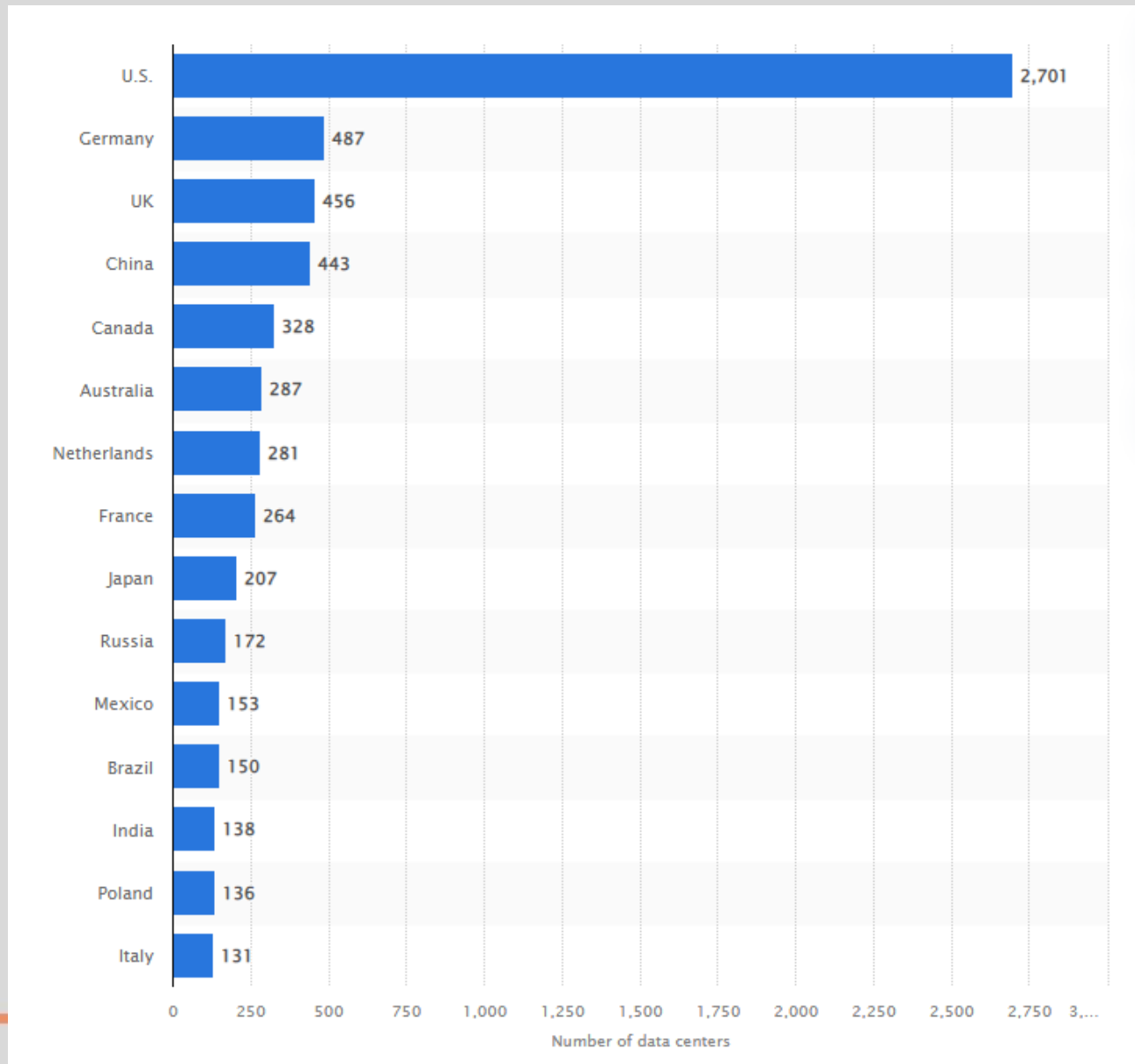
Data Center

58 regions
worldwide

140 available in
140 countries

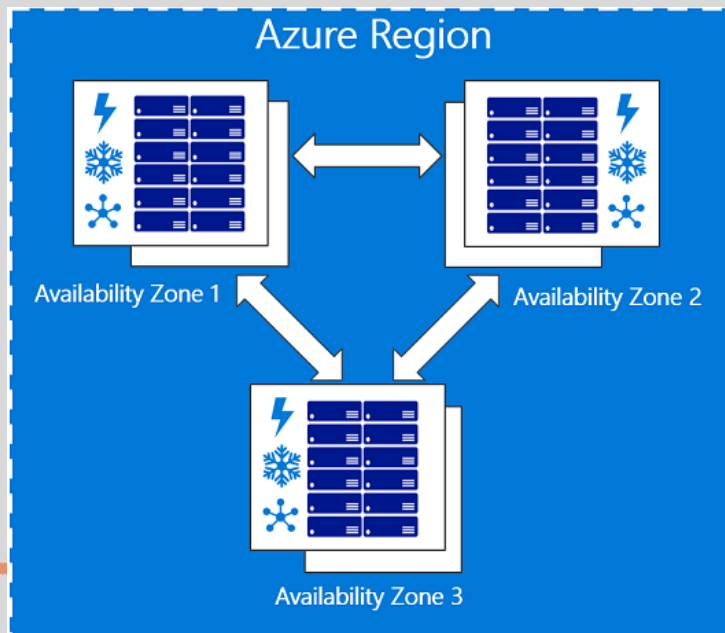


Number of data centers (2022)



Availability Zones/Fault Domains

- Unique physical locations within a region
- Each zone is made up of one or more datacenters equipped with independent power, cooling, and networking
 - To ensure resiliency, there's a minimum of three separate zones in all enabled regions
 - The physical separation of Availability Zones within a region protects applications and data from datacenter failures



- Protecting application and data from datacenter failures
- Protecting from single-point-of-failure
- Increase SLA (high-availability)

«Contenitore» logico

- Molti cloud hanno un concetto di virtual datacenter (o SDDC)
 - O semplicemente un «contenitore» logico
- Diversi nomi
 - AWS
 - Virtual Private Cloud (VPC)
 - Azure
 - Azure Resource Group e vNET
 - GCP
 - Project e Virtual Private Cloud

Caratteristiche del cloud

- Affidabilità (Reliability), stabilità (Stability), disponibilità (Availability) e ripristinabilità (Recoverability)
 - Livelli di servizio ben definiti
 - Contratti con SLA dichiarati
 - Cloud Design Pattern sia per disponibilità che scalabilità
- Scalabilità (Scalability)
 - PaaS e IaaS sia in modalità scale-in che scale-out
 - Nel SaaS la scalabilità è gestita dal provider
- Prestazioni (Performance)
 - Tempi di attesa non devono essere eccessivi
 - Importanza di region vicine o di servizi edge
- Manutenibilità (Manageability) e semplicità
 - Deve essere semplice ma anche efficace
 - Tutto automatizzabile (ad esempio con «cloud shell»)

Auto-scaling

- Particolarità dei servizi PaaS e IaaS
 - <https://aws.amazon.com/it/autoscaling/>
 - <https://learn.microsoft.com/it-it/azure/architecture/best-practices/auto-scaling>
 - <https://cloud.google.com/compute/docs/autoscaler?hl=it>
- Diverse istanze in parallelo
 - Di fatto un cluster LB gestito
 - Soglie di crescita e soglie di decrescita
 - Ad esempio sull'uso della CPU
 - Soglie massime per limitare i costi
 - Max numero di istanze

Cloud «cosa»?

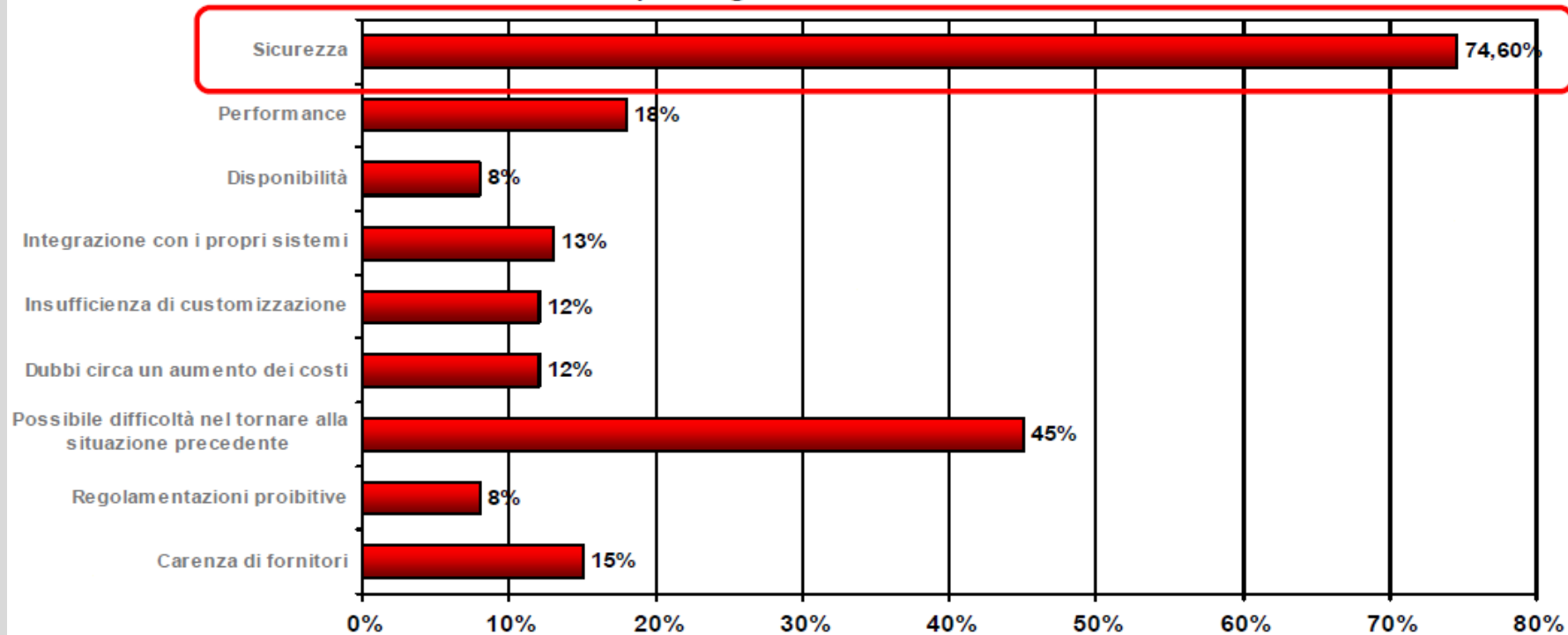
- Cloud OS
- Cloud Management Platform
- Cloud Resource Manager
- Cloud Orchestrator
- Cloud Monitor
- Cloud Cost Analyzer

Cloud e normative

- Norme relative a privacy e trattamento dati
 - (EU) GDPR
 - (USA) Data Cloud Act
- Restrizioni specifiche su dove devono risiedere i dati
 - Suolo nazionale
 - Comunità europea
 - ...
- Garante per la protezione dei dati personali
 - Cloud computing: indicazioni per l'uso consapevole dei servizi
- Standard
 - IEEE P2301
 - Guide for Cloud Portability and Interoperability Profiles (CPIP)
 - IEEE P2302 Draft
 - Draft Standard for Intercloud
 - Interoperability and Federation (SIIF)

Rischi correlati?

Gli utenti percepiscono gli aspetti di sicurezza come principale rischio nell'adozione del Cloud Computing.



Shadow IT

61% of SMBs agree or strongly agree that their company encourages and rewards collaboration

70% of small firms agree or strongly agree that their communication and collaboration solutions help improve productivity

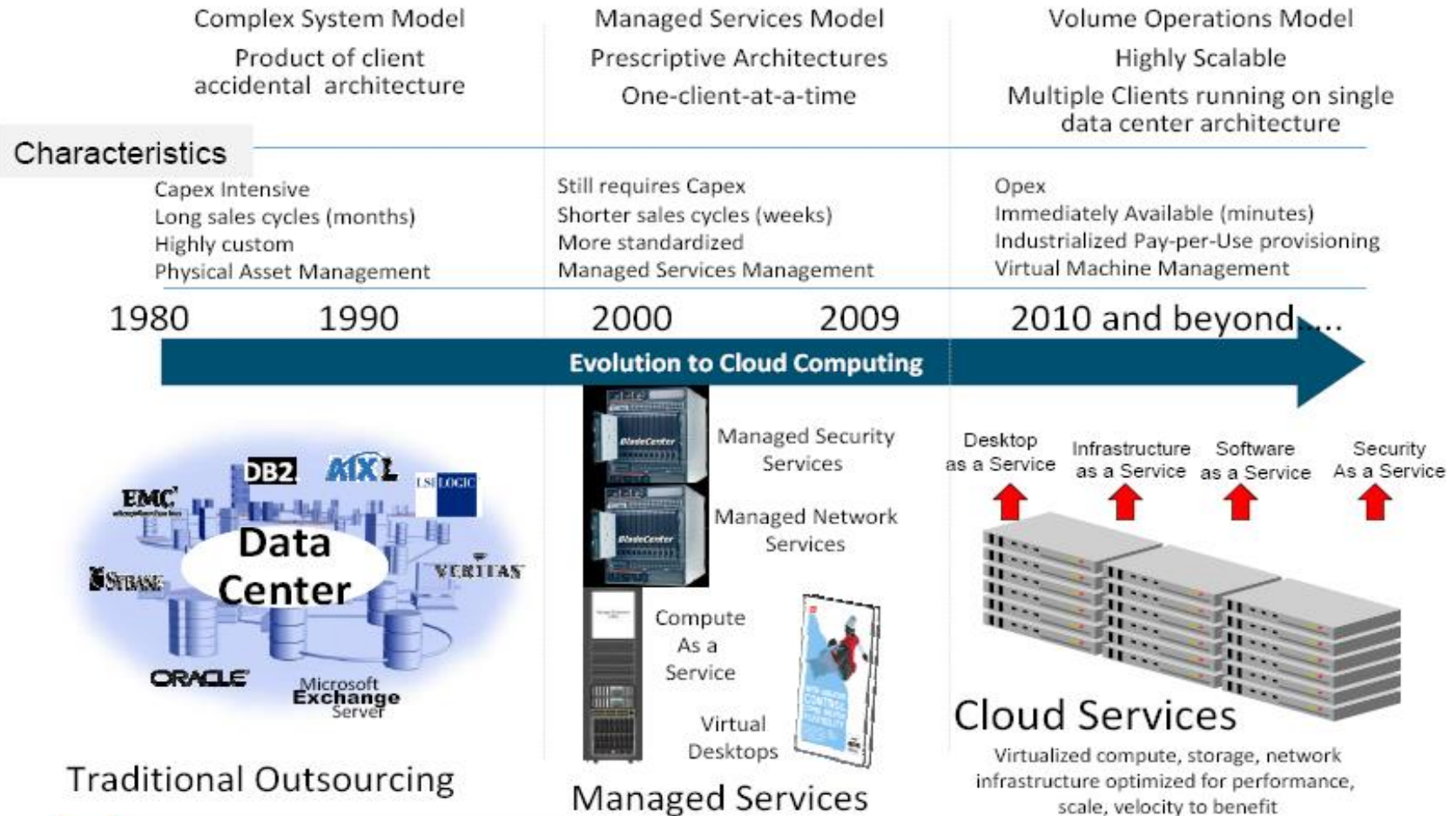
Only 38% of technology purchases are now managed, defined and controlled by business leaders

Shadow IT is evolving

“...it’s easy for employees to use “unsanctioned” collaboration solutions, which can create support problems and security risks.”

Source: [Make the Best of Shadow IT](#)
[Trends in SMB Collaboration, Communication, and Mobility: What's Your Strategy?](#)

L'evoluzione della specie



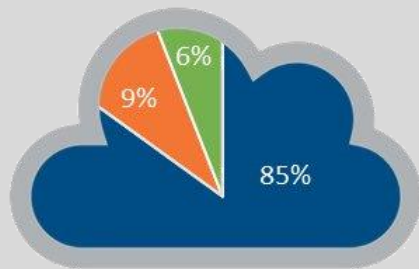
IT vs Cloud

Spesa Infrastruttura IT vs Cloud

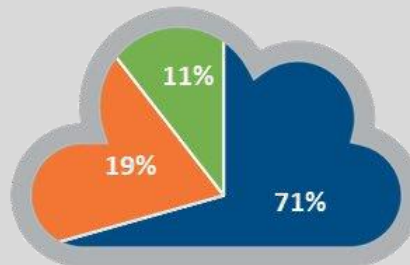
Traditional IT

Private Cloud

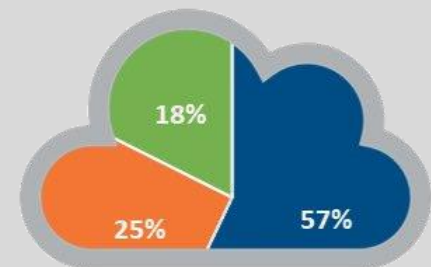
Public Cloud



2013



2017



2021

Digital transformation



Traditional
applications

Cloud-native
applications

IT Investments \$

IT transformation

Optimize IT for the business

Digital transformation

IT is the business

2000

2015

2030

Verso il cloud

Traditional

IT-centric and project-focused

Siloed organization

Static environment

Heavy governance, or nothing at all

Manual IT service development and delivery

Traditional cost management



Cloud

Customer-focused, service-oriented culture

Cross-functional and multi-disciplinary teams

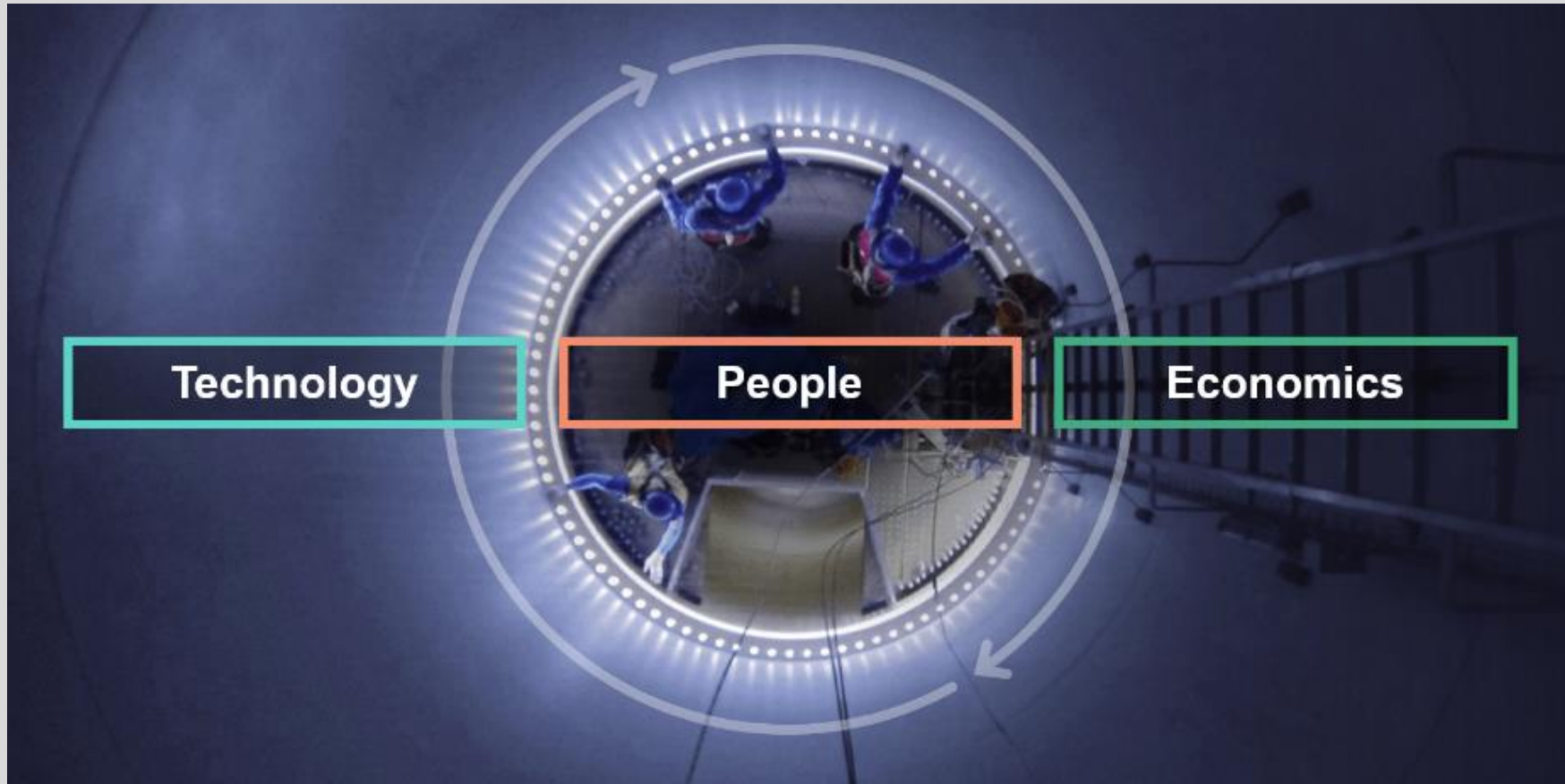
Dynamic environment, where change is the norm

Adopting a model for cloud operations and management

Automated delivery of IT services from development to consumption

Cross cloud, service-based cost management, with a pay-as-you-consume model

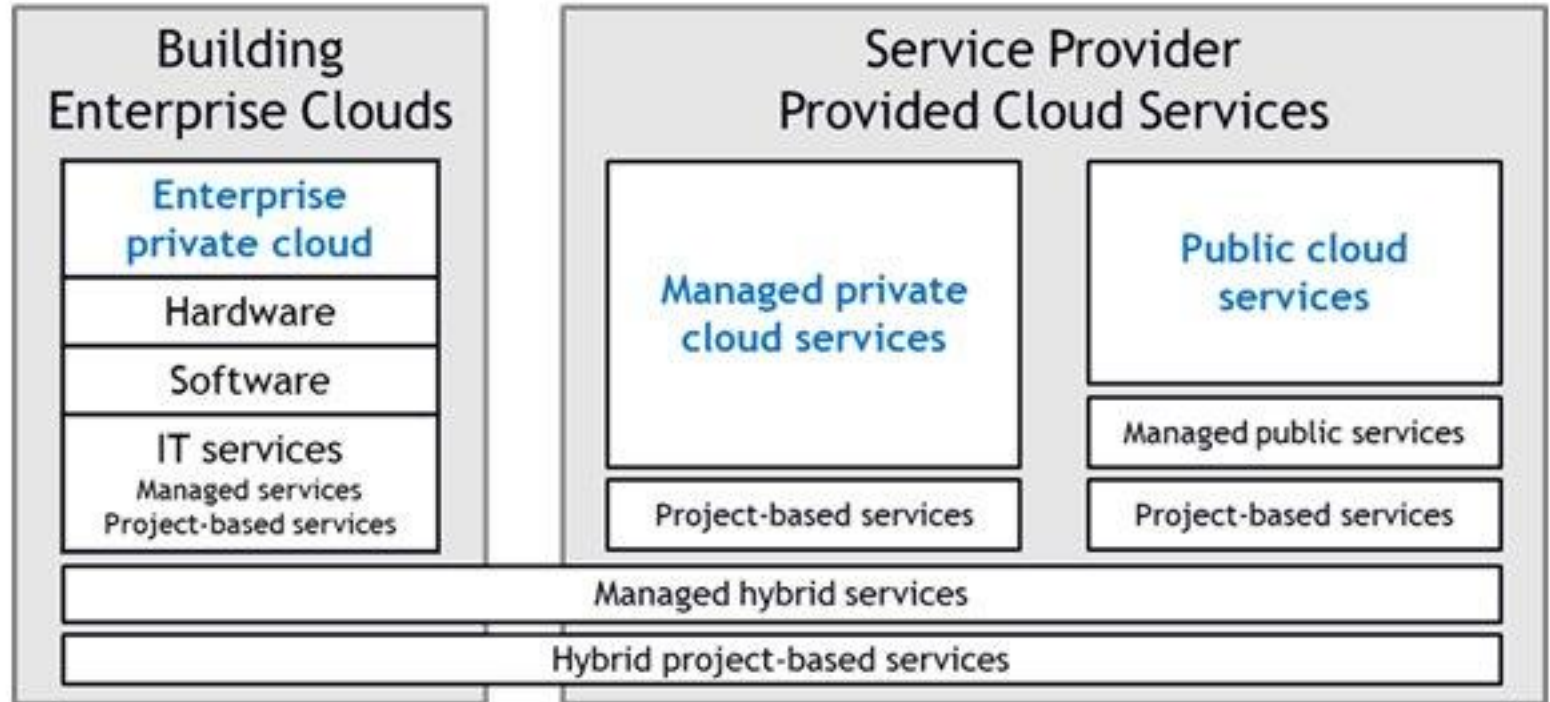
Non solo tecnologia



Verso i Private Cloud

- Evoluzione dell'infrastruttura IT aziendale
 - «Servizio centrica»
 - l'infrastruttura esiste per erogare servizi al business in modo trasparente rispetto alla reale implementazione
 - Agile e dinamica
 - in grado cioè di adeguarsi in modo rapido alle necessità aziendali implementando nuovi servizi o scalando quelli esistenti verso l'alto o verso il basso
 - Affidabile
 - il business deve poter contare su servizi standard, predicibili e resilienti
 - Efficiente
 - la maggior parte delle operazioni giornalieri sono automatizzate in modo da raggiungere il maggior livello possibile di auto-gestione (es. distribuzione o ritiro di nuovi servizi al variare del carico di lavoro)
 - A costi trasparenti
 - le unità di business, e l'IT stesso, devono poter conoscere il in modo chiaro il costo dei servizi richiesti all'IT in modo da poter prevedere la spesa necessaria a sostenerli e decidere se usufruire di servizi interni o di analoghi servizi esterni

Verso il cloud ibrido

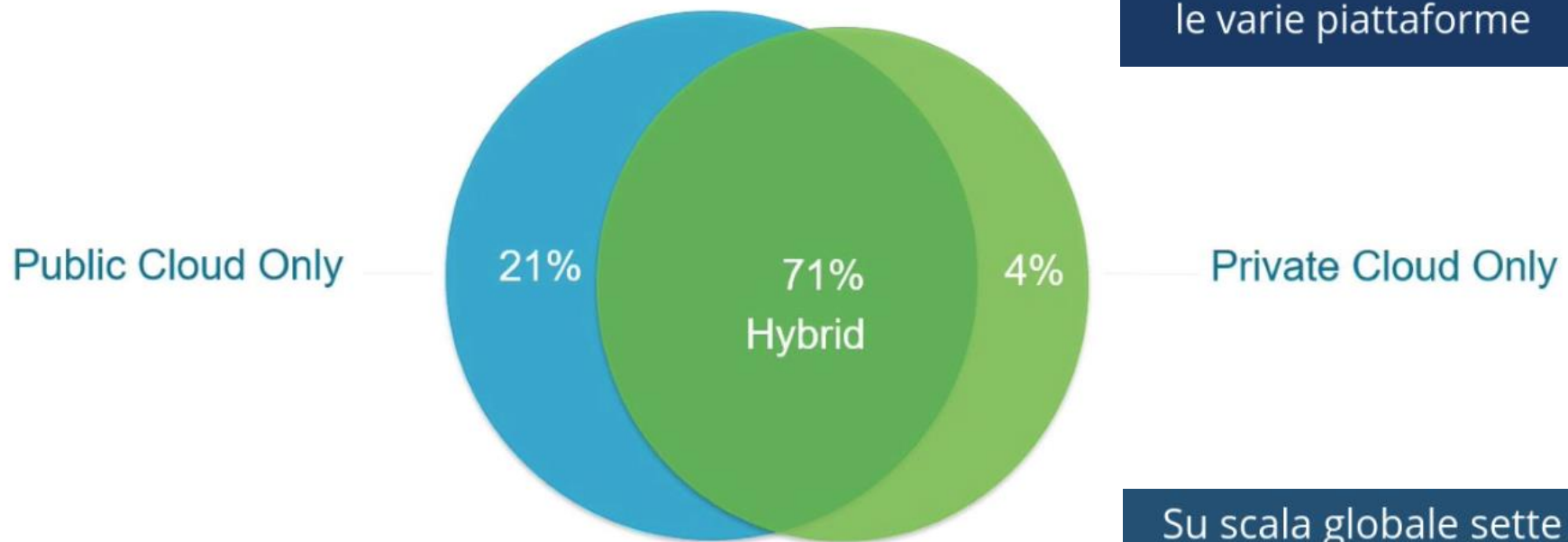


Note: For more details, see *IDC's Worldwide Managed Cloud Services Taxonomy*,
Source: IDC, 2017

Hybrid Cloud

Hybrid Cloud is the Preferred Enterprise Strategy

Combining the best of both worlds into a single strategy



Source: RightScale 2018 State of the Cloud Report

La modalità hybrid cloud aggiunge anche l'integrazione e l'orchestrazione tra le varie piattaforme

Su scala globale sette aziende su dieci adottano il multcloud (in Europa il 67%), solo una su dieci in modalità hybrid cloud

Private/Hybrid Cloud Solutions

- Soluzioni proprietarie
 - VMware SDDC
 - VMware Cloud Foundation
 - VMware vRealize Suite
 - Microsoft Azure Stack / Azure Arc
 - AWS Outposts
 - GCP Anthos
- Soluzioni OpenSource
 - OpenStack
 - OpenNebula

Multi-cloud

- Hybrid always includes private and public, while multi-cloud always includes multiple public clouds but can also incorporate physical and virtual infrastructure (including private clouds)
- Unlike a multi-cloud model, in which different clouds are used for different tasks, the components of a hybrid cloud typically work together
 - As a result, data and processes tend to intermingle and intersect in a hybrid environment, while in a multi-cloud situation, usage typically remains in its “own” cloud’s silo

E' multcloud un ambiente che si basa su due o più cloud (public o private) di provider IaaS, PaaS e SaaS differenti

Multi-cloud in Italy

Multicloud is a reality in Italy, however orchestration and automation are complex



Fonte: Survey IDC per Assintel Report, 2020

What about multi-cloud automation and orchestration processes?

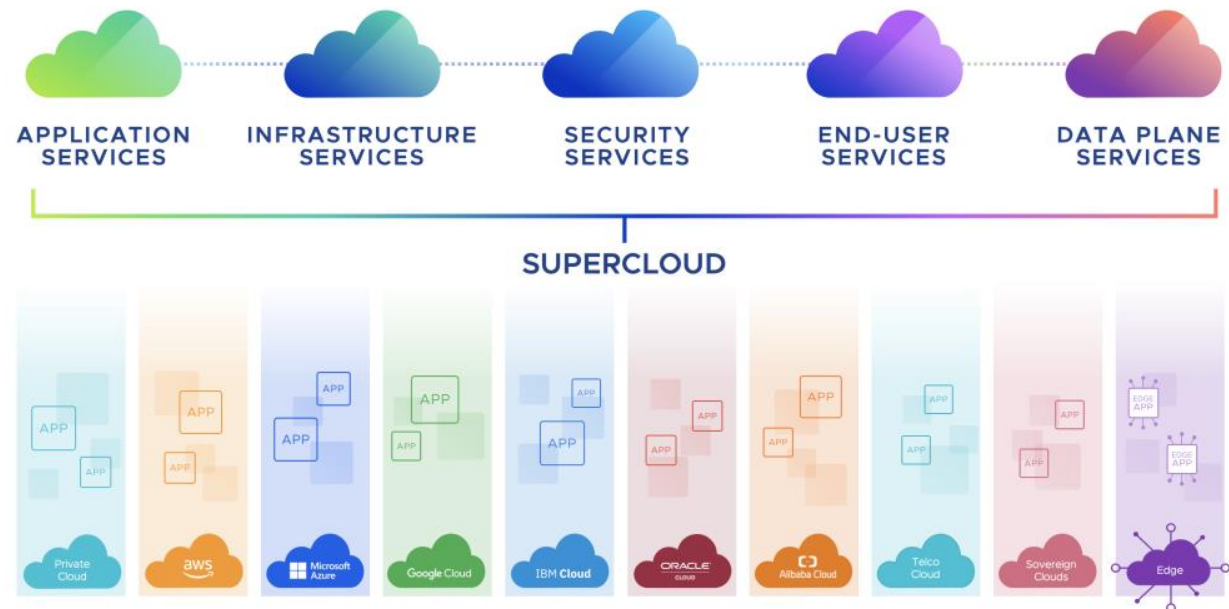
Italy is slightly behind Western European average

(in terms of % of companies positioned at the most advanced stages of multi-cloud automation and orchestration processes)

Source: IDC Europe, DevOps & Developers Survey, 2019

Supercloud

- Multi-cloud is really not a strategy but rather a symptom of multivendor
- Supercloud
 - abstraction layer that resides above and across infrastructure
 - cloud architecture that enables application migration as a service across different availability zones or cloud providers
 - goes beyond running services in native mode on each individual cloud



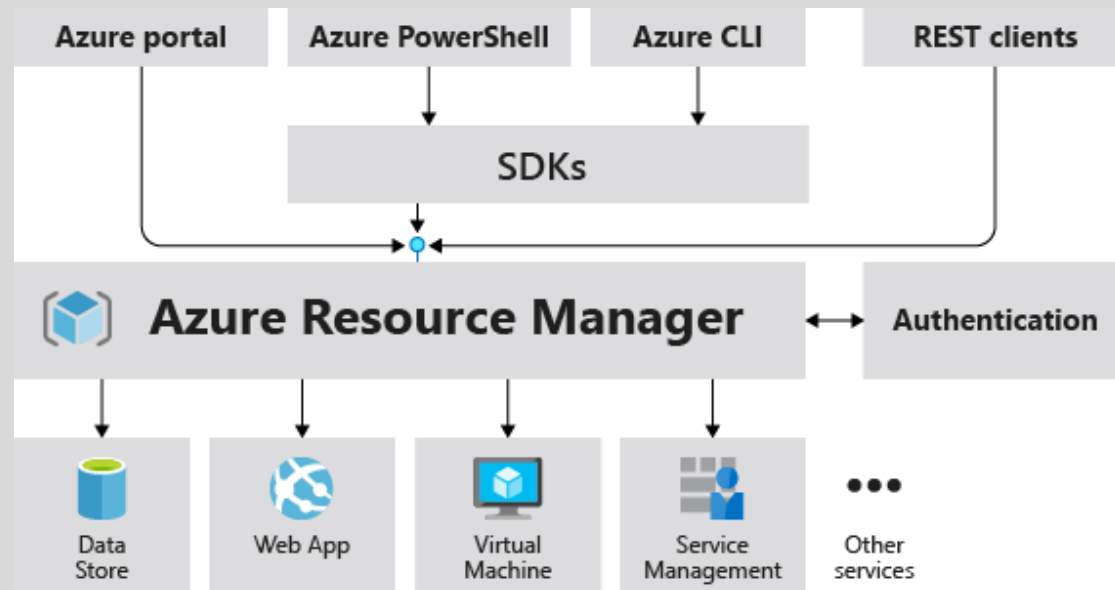
Cloud Management Platform

- «Portale» delle risorse cloud
 - SaaS nel caso del public cloud
 - Ma limitato alle risorse del provider
 - Necessario per implementare private cloud
 - Utile per hybrid cloud e multi-cloud
- Diverse soluzioni
 - Ad esempio
 - OpenStack
 - VMware vRealize Automation
 - Microsoft Azure Arc
- Diverse funzionalità richieste
 - Fondamentale l'integrazione con altri prodotti
- Da completare con la gestione/analisi costi

Azure Resource Management

- Azure Resource Manager is the deployment and management service for Azure
 - It provides a management layer that enables you to create, update, and delete resources in your Azure account
 - You use management features, like access control, locks, and tags, to secure and organize your resources after deployment

- ▶ Manage your infrastructure through declarative templates rather than scripts.
- ▶ Deploy, manage, and monitor all the resources for your solution as a group, rather than handling these resources individually.
- ▶ Redeploy your solution throughout the development lifecycle and have confidence your resources are deployed in a consistent state.
- ▶ Define the dependencies between resources so they're deployed in the correct order.



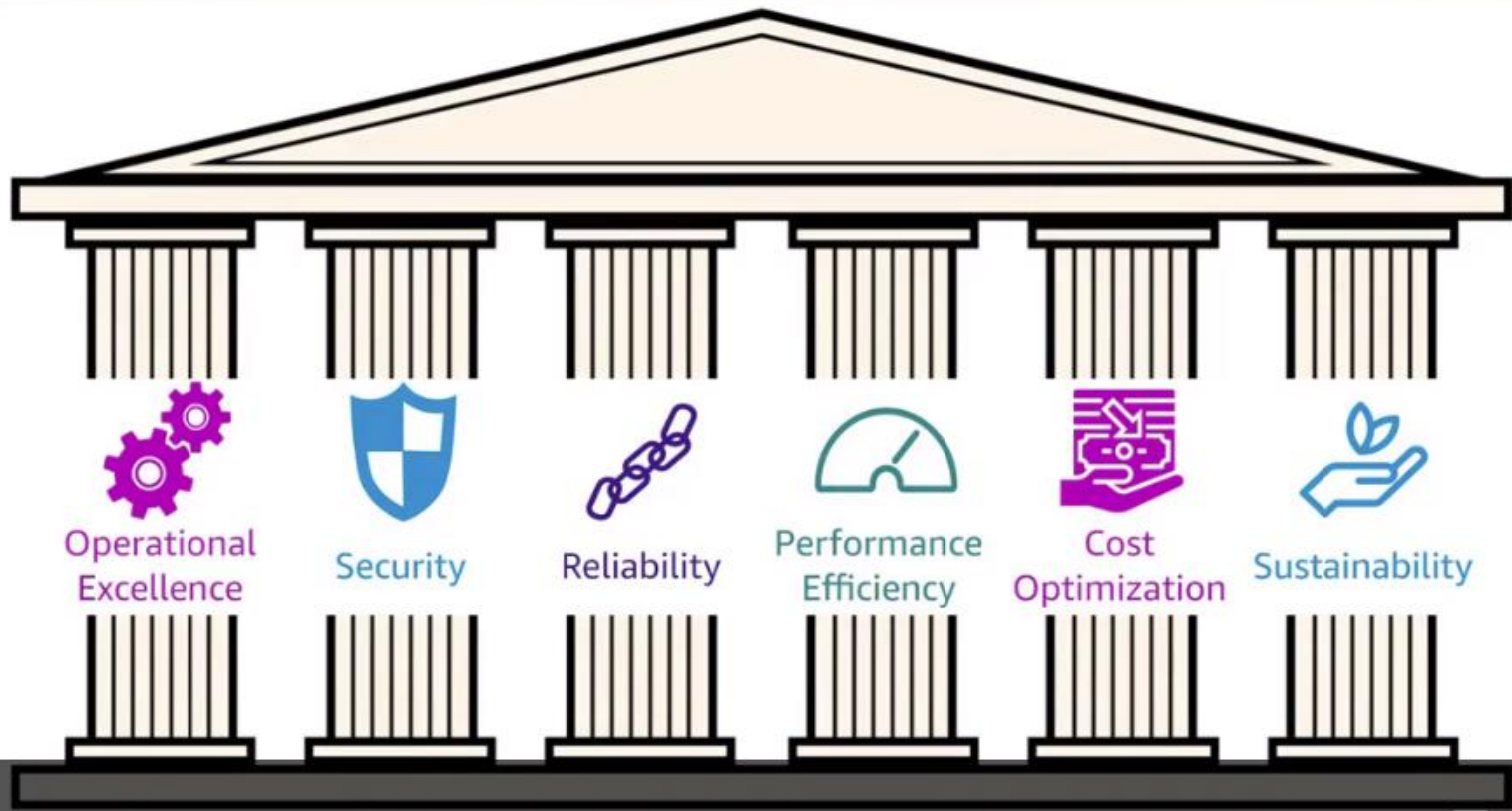
Migrare verso il cloud

- Diversi problemi da risolvere
 - Anche di tipo organizzativo / legislativo / compliance /....
 - Complessità aggiuntiva per modelli multi-cloud e/o hybrid-cloud
- Come portare i servizi su una piattaforma di cloud computing?
 - Lift & Shift
 - Sposto «as-is»
 - Cloud Native
 - Si riprogetta per sfruttare il cloud al meglio
- A volte si intende migrare verso il modello cloud computing e non verso un cloud

Cloud architecture best practices

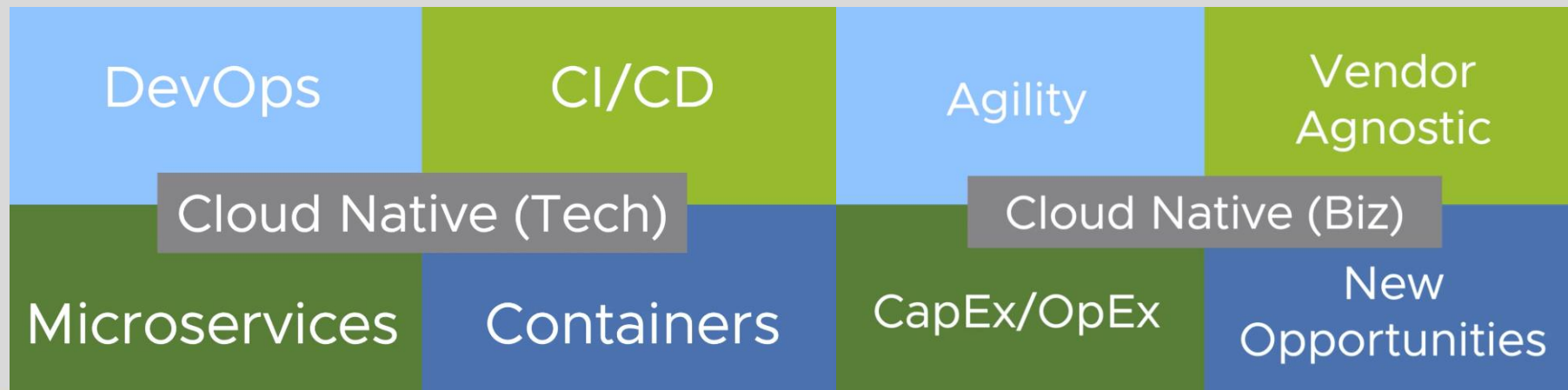
1. Design for failure and nothing fails
2. Build security in every layer
3. Leverage different storage options
4. Implement elasticity
5. Think parallel
6. Loose coupling sets you free
7. Don't fear constraints

The Six Pillars



Cloud Native Application

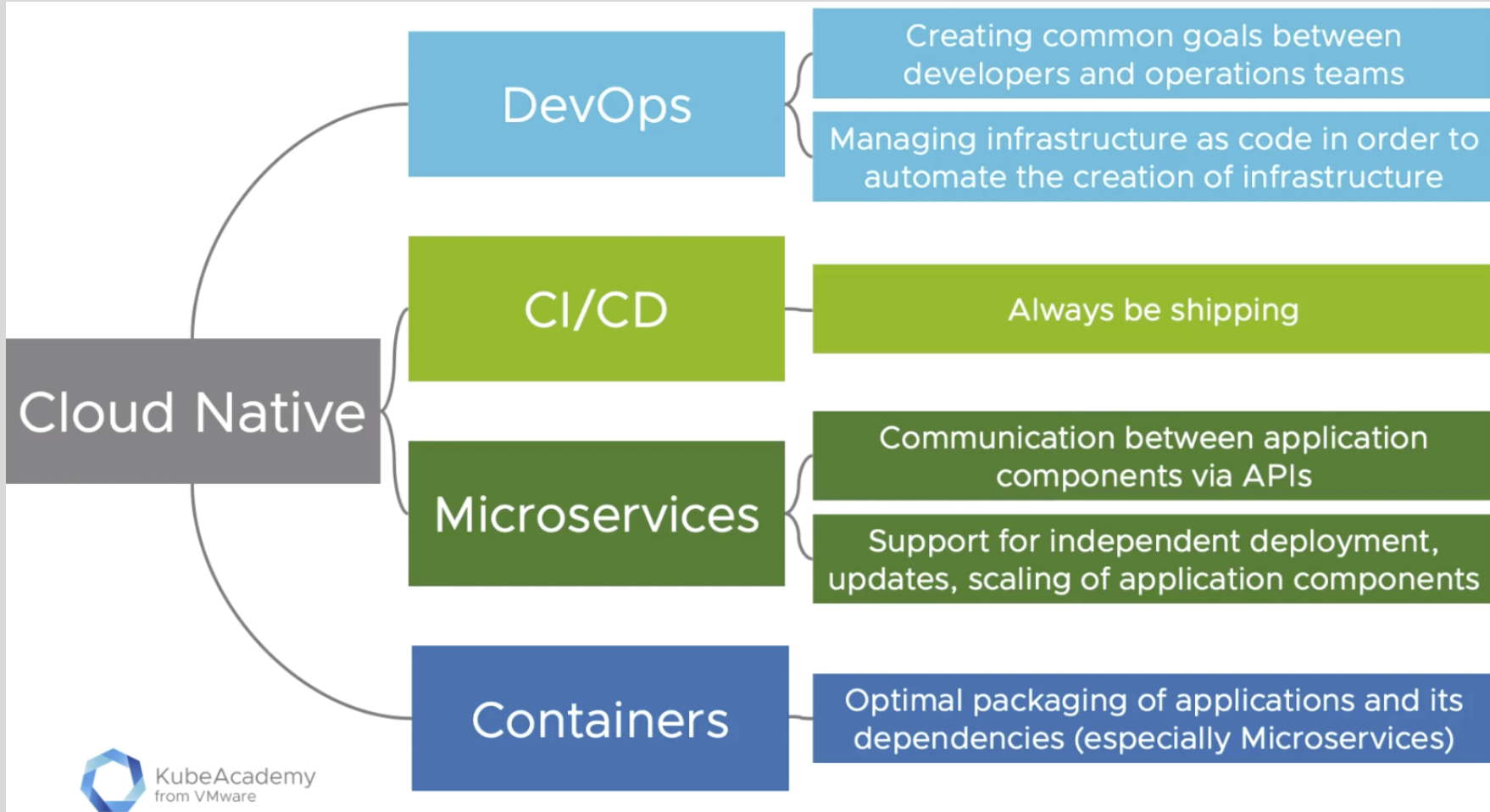
- Modern Application
- Applicazioni pensate per lavorare secondo il modello del cloud
 - Che sfruttano i vantaggi del cloud
- 12-Factor App
 - <https://12factor.net/it/>



Cloud Native Application

- Requisiti per una CNA
 - Scale on demand
 - Self-heal
 - Support zero-downtime rolling updates
 - Run anywhere
- A cloud-native application is not an application that only runs in the public cloud
 - *taking advantage of the unique capabilities of the cloud* – Joe Beda

Cloud Native Application



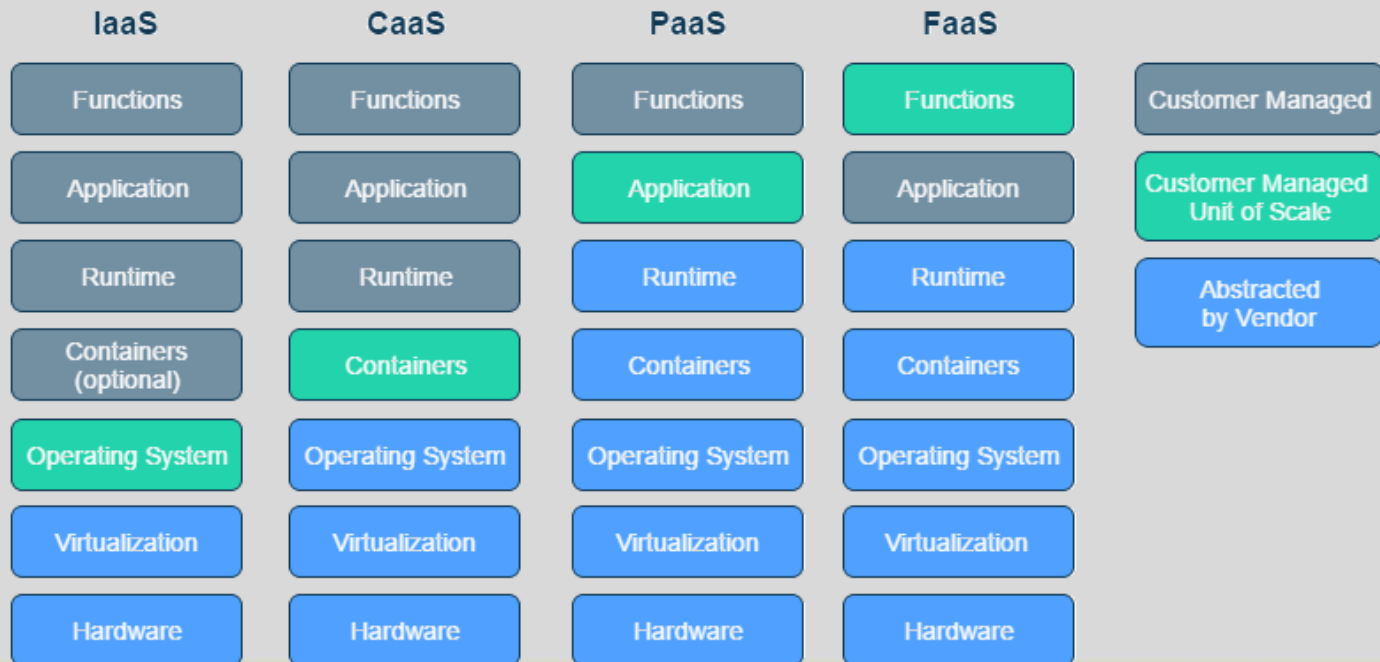
Pets vs. Cattle

Pets vs. Cattle



Da server a serverless

- Molti servizi cloud sono di tipo client-server
- Ma si stanno diffondendo soluzioni serverless
 - Function as a Service (FaaS)
 - Es: AWS Lambda

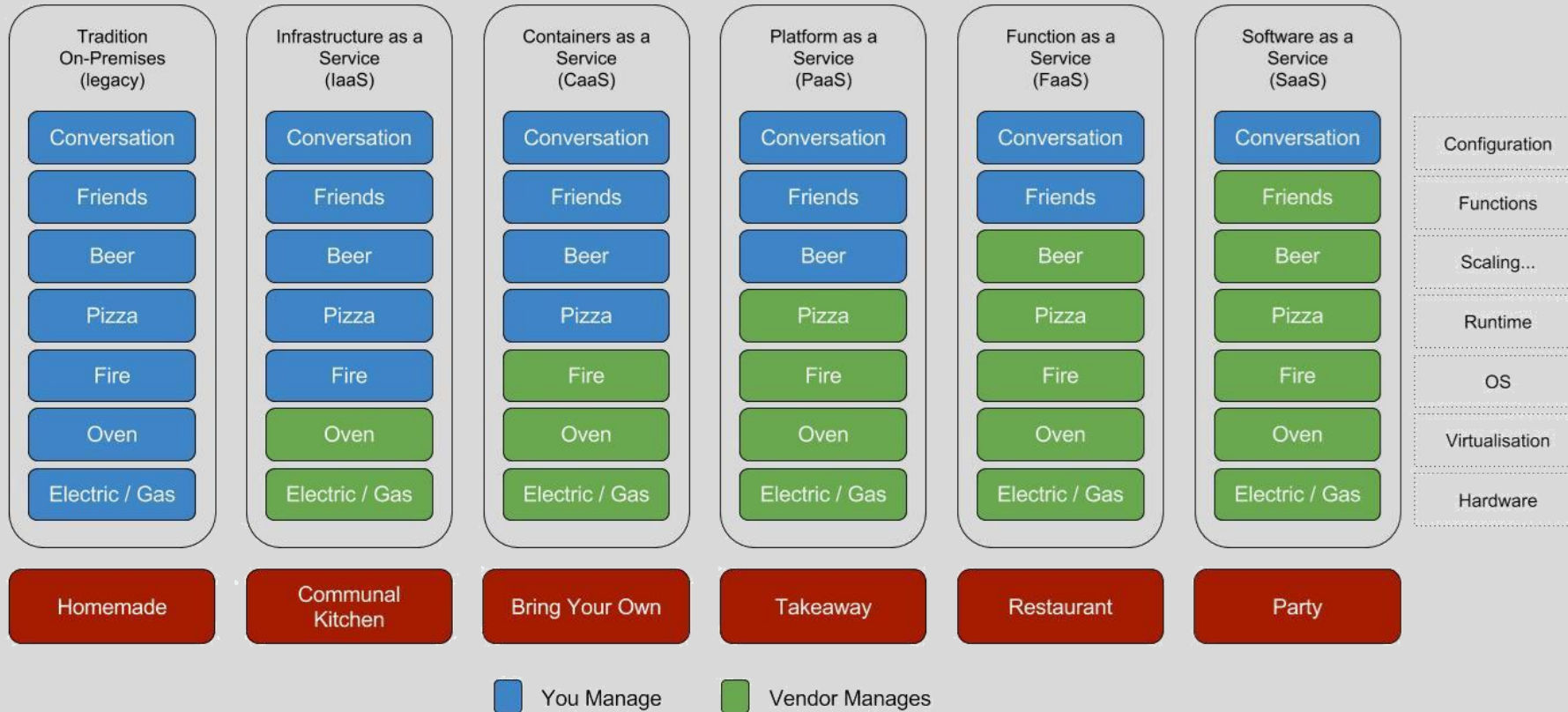


Pizza as a Service 2.0



Pizza as a Service 2.0

<http://www.paulkerrison.co.uk>

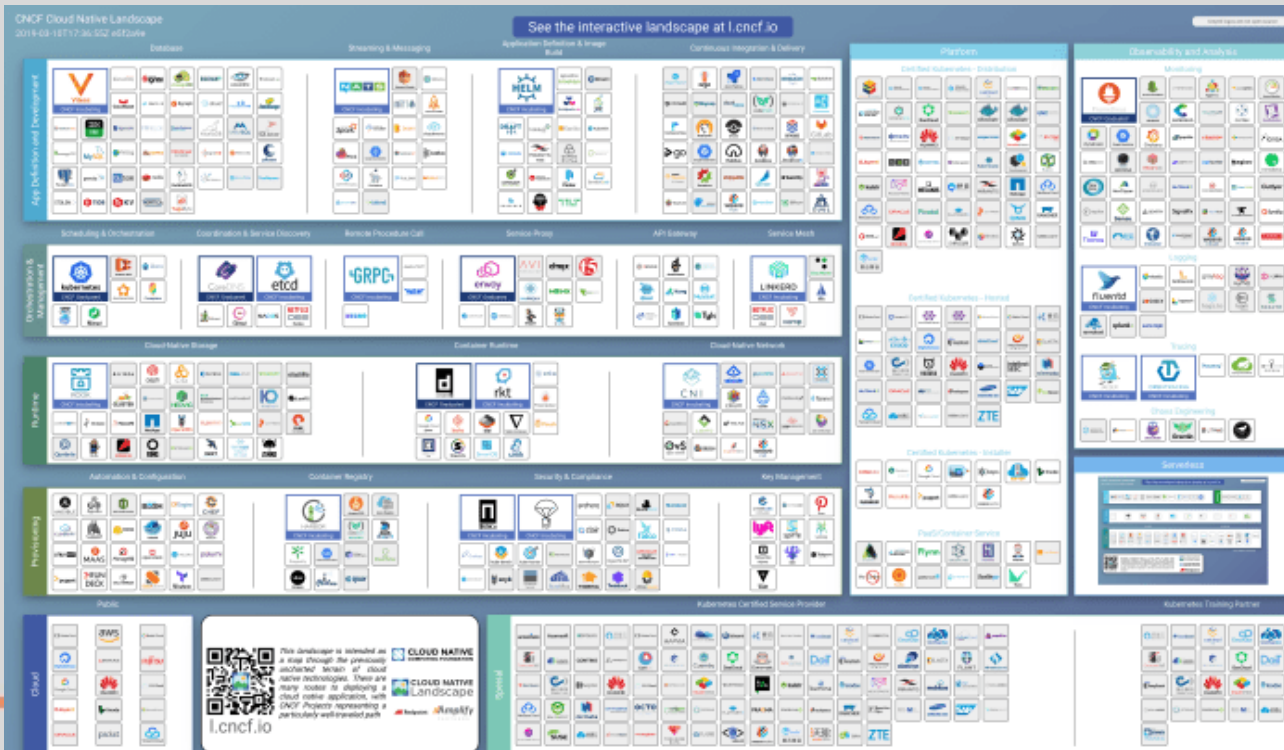


Pets vs. Cattle vs. ...

- Pets: the bare metal servers and virtual machines
- Cattle: the sort of instances you run on public clouds
- Chickens: an analogy for containers
- Insects: An analogy for serverless
- Snowflakes

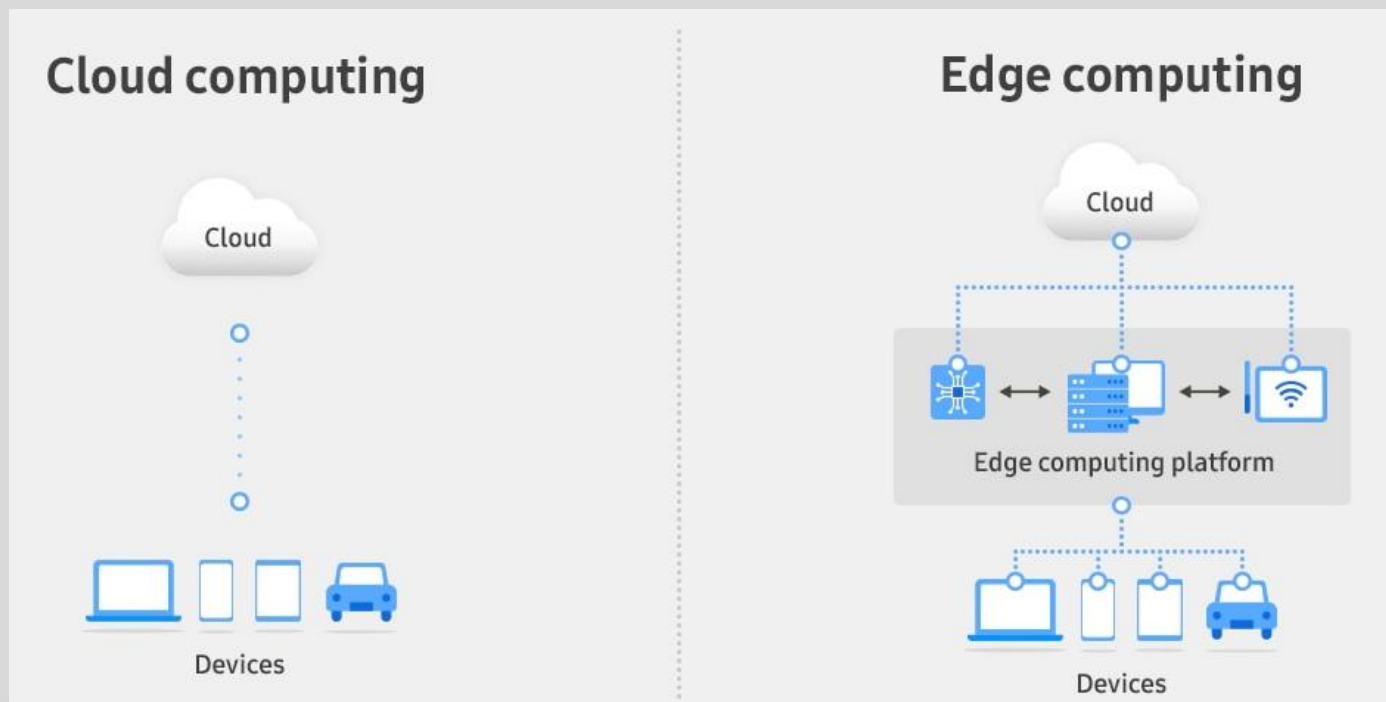
CNCF

- Cloud Native Computing Foundation (CNCF) is a Linux Foundation project that was founded in 2015 to help advance container technology and align the tech industry around its evolution
 - <https://www.cncf.io/>
- CNCF Cloud Native Interactive Landscape
 - <https://landscape.cncf.io/>



Edge Computing

- Modello di calcolo distribuito nel quale l'elaborazione dei dati avviene il più vicino possibile a dove i dati vengono generati
 - migliorando i tempi di risposta e risparmiando sulla larghezza di banda
 - Evoluzione delle Content Delivery Network (fine degli anni '90)
 - Utilizzato soprattutto nell'IoT



Edge Computing



Fog Computing

- Fog computing, also called Edge Computing, is intended for distributed computing where numerous "peripheral" devices connect to a cloud
- The OpenFog Consortium
 - founded by Cisco Systems, Intel, Microsoft, Princeton University, Dell, and ARM Holdings in 2015
 - merged with the Industrial Internet Consortium on January 31, 2019



Cloud e sicurezza

Statistics

38% increase in Cybercrime-as-a-Service targeting business email observed between 2019 and 2022

35 million business email compromise attempts, with an average of 156,000 attempts daily, detected and investigated by Microsoft Threat Intelligence between April 2022 and April 2023

Ransomware attacks occur roughly every 11 seconds, and 94% of them now target backups

21% Share of incidents that saw backdoors deployed

62% Percentage of phishing attacks using spear phishing attachments

Microsoft¹

NetworkComputing²

IBM³

1. <https://shorturl.at/cpxN6>

2. <https://www.networkcomputing.com/network-security/structured-success-4-architectural-pillars-cyber-resilience>

3. <https://www.ibm.com/downloads/cas/DB4GL8YM>

Documenti utili

- Cloud services for dummies
 - <https://www.ibm.com/cloud-computing/files/cloud-for-dummies.pdf>
- The Definitive Guide to Cloud Computing
 - http://eddiejackson.net/web_documents/The_Definitive_Guide_to_Cloud_Computing.pdf
- AWS for beginners
 - <https://douran.academy/wp-content/uploads/ebooks/aws-for-beginners.pdf>
- Azure for Architects
 - <https://info.microsoft.com/ww-landing-azure-for-architects.html>
- Corsi gratuiti
 - Azure: <https://learn.microsoft.com/en-us/training/azure/>
 - AWS: <https://aws.amazon.com/it/training/learn-about/cloud-practitioner>